

COMPLETE PROJECT REPORT

PROJECT SENTINEL

Autonomous AI-Powered Payment Risk Intelligence & Protection Platform

Combining Machine Learning, Graph Intelligence, Agentic AI and PayPal Payment Infrastructure

PREPARED FOR

PayPal AI Hackathon 2026

PREPARED BY

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Disclaimer: Sentinel is a hackathon prototype. It runs on PayPal Sandbox with synthetic data only and is not an official, deployed or endorsed PayPal fraud-prevention product. It does not access PayPal's internal risk systems. Model results describe synthetic data and are not evidence of real-world performance.

About this document

The complete technical and non-technical report (Parts 1 to 18): problem, solution, architecture, implementation, PayPal integration, testing, limitations and conclusion. Parts 1 to 4 are written for non-technical readers. The interview question bank and personal pitches are in the separate Viva and Interview Guide.

Item	Detail
Project	PROJECT SENTINEL: Autonomous AI-Powered Payment Risk Intelligence & Protection Platform
Prepared for	PayPal AI Hackathon 2026
Prepared by	Tanisha Dey
Source of truth	The repository (backend/, frontend/, docs/). Every metric and test result was taken from the running system or its test suite.
Repository	github.com/Tanisha-dey18/hackathon
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> **Disclaimer.** Sentinel is a hackathon prototype using PayPal Sandbox and synthetic data. It is not an officially deployed PayPal fraud prevention product and has no access to PayPal's internal systems. Features that are designed but not implemented are labelled **Future Scope**, and integrations that are implemented but were not executed live are labelled as such.

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PART 1

Executive Overview

1.1 What is Sentinel?

Sentinel is a payment protection layer that sits between a customer approving a payment in PayPal and the moment the merchant actually collects the money. It looks at each payment, decides how risky it is, explains why, and only lets PayPal collect the funds when that is safe or when a human reviewer has approved it.

The project has two faces:

- **SENTINEL MERCHANT**: a demo online shop where a shopper picks products and pays with PayPal (Sandbox).
- **The Sentinel console**: a dashboard used by risk analysts and reviewers to watch transactions, see why something was flagged, explore relationships between accounts, run an AI investigation and approve or reject held payments.

> In one sentence: Sentinel decides, with evidence, whether money should move, and keeps a human in charge of the decisions that cannot be undone.

1.2 Why was Sentinel created?

Online payments are fast, and fraud is fast too. Many fraud checks look at one payment at a time and return a score with no explanation. That leaves two problems. First, organised fraud is often invisible at the level of a single payment: ten accounts that share one phone and send money to the same wallet each look normal on their own. Second, when a payment is flagged, someone still has to work out why, which takes time while the customer waits.

Sentinel was built for the PayPal AI Hackathon 2026 to show how payment infrastructure, machine learning, graph analysis and an AI investigator can work together in one coherent workflow.

1.3 What problem does it solve?

Inside this prototype, Sentinel solves a specific, well-defined problem: "*Before PayPal captures an approved payment, decide whether it is safe to capture, explain the decision, and make sure risky payments cannot be captured without an authenticated human approval.*"

1.4 Who would benefit?

Who	Benefit
Merchants	Fewer fraudulent captures and the chargebacks that follow; fewer genuine customers wrongly declined, because flagged payments are reviewed rather than auto-rejected.
Risk reviewers and analysts	A ready-made case file: transaction details, customer history, model explanation, relationship graph and an evidence-cited recommendation.
Genuine customers	Normal purchases go through automatically; unusual ones are checked by a person instead of being silently blocked.
Engineering and compliance teams	Every decision is recorded in a tamper-evident audit log.

1.5 How does it work? (simple version)

1. A shopper adds items to the cart and clicks the PayPal button.
2. Sentinel's server creates a real PayPal order with prices taken from its own catalogue.

3. The shopper approves the payment in the PayPal window.
4. Before any money is collected, Sentinel asks PayPal for the order, checks the amount and that it belongs to this shopper, and then scores the risk.
5. Low-risk payments are captured automatically. Anything else is **held**, which means PayPal has the buyer's approval but no money has been collected.
6. For high-risk payments an AI investigator collects evidence and writes a report with a recommendation.
7. A reviewer signs in again, reads the evidence and approves, rejects or escalates. Only an approval can trigger the PayPal capture.

1.6 What makes it different?

- **Risk is enforced at the exact point where money moves.** The capture step itself checks that the payment was either LOW risk or approved by a reviewer.
- **Three kinds of intelligence, one decision.** A supervised ML model, an unsupervised anomaly detector and a relationship graph each contribute, and the policy explains which one raised the risk.
- **An AI investigator that cannot make things up.** Every finding in its report must cite an evidence item produced by a tool; anything else is removed automatically.
- **A clear separation of truth.** PayPal's payment status is only ever copied from PayPal. Sentinel's risk decision is stored separately, so a held payment is never shown as paid.

1.7 Why PayPal integration is important

Without a real payment system, a fraud model is just a classifier. With PayPal, Sentinel controls a real lifecycle: order creation, buyer approval and capture. The risk decision has a concrete consequence (capture or do not capture), and PayPal's own idempotency and webhook signature verification are part of the safety design.

1.8 Where AI fits

AI component	What it does in Sentinel
Machine learning (Random Forest, XGBoost, Logistic Regression)	Estimates the probability that a payment is fraudulent from 26 behavioural features.
Anomaly detection (Isolation Forest)	Measures how unusual a payment is compared with all training payments, without using labels.
Explainable AI (SHAP)	Shows which features pushed the score up or down for each individual payment.
Graph intelligence (NetworkX)	Finds shared devices, money loops and clusters of linked accounts.
Agentic AI (LangGraph + Groq)	Runs a multi-step investigation with tools and writes an evidence-backed report. A rule-based fallback is used, and labelled, when Groq is not configured.

1.9 The one-minute explanation

"Sentinel protects PayPal checkouts. When a shopper pays, our server creates a real PayPal Sandbox order. After the shopper approves it, and before we capture any money, Sentinel scores the payment with a machine-learning model, checks how unusual it is, and looks at a graph of which accounts share devices or send money to the same wallets. Normal payments are captured straight away. Suspicious ones are held: PayPal shows the order as approved, but nothing is collected. An AI agent then gathers evidence with seven tools and writes a report where every claim points to a piece of evidence. A reviewer re-enters their password and approves, rejects or escalates, and only an approval lets PayPal capture the money. Everything is recorded in a hash-chained audit log. It all runs on synthetic data and PayPal Sandbox, so no real customers or money are involved."

PART 2

Problem Statement

2.1 The broader problem

Digital payments keep growing, and so does the attack surface. The patterns below are widely discussed in the payments industry; they are described here as general context, not as statistics measured by this project.

- **Suspicious payment behaviour**: purchases far above a customer's normal spending, at unusual hours, or in quick bursts.
- **Account takeover (ATO)**: a criminal logs into a real customer's account from a new device, often from a far-away location, and buys high-value, resellable goods.
- **Card testing**: many tiny failed payments to check which stolen details work, followed by a large purchase.
- **Coordinated fraud networks**: many accounts controlled by the same group share devices and move money to the same "mule" destinations, or pass money around in circles.
- **Rule-based limits**: fixed rules (for example "block anything above $1,000") are easy to explain but easy to evade, and they block many genuine customers.
- **False positives**: every genuine customer wrongly declined is lost revenue and a damaged relationship.
- **Detection is not enough**: a score with no explanation still leaves an analyst to rebuild the story by hand.
- **Explainability matters**: payment decisions affect real people; reviewers, auditors and customers need to understand why a payment was held.

2.2 A realistic hypothetical example

Cara's account has a short purchase history. On a Tuesday afternoon she buys a $500 digital gift card. On its own that is not alarming: the amount is about 1.5 times her average, and she has used this phone before. But that phone is also used by three other accounts, and those accounts have repeatedly sent money to two external wallets that are known to have received funds linked to chargebacks. A model that only looks at Cara's payment sees an ordinary purchase. A system that looks at the relationships sees a likely fraud ring. This exact situation is reproduced as Demo Scenario C (Part 14), where the ML model scored the payment at about 0.33 but the graph policy raised it to HIGH.

2.3 Industry challenges versus what this prototype solves

General industry challenge	What Sentinel's prototype actually does
Fraud at very large scale	Not addressed. The prototype processes a few thousand synthetic transactions in one process.

General industry challenge	What Sentinel's prototype actually does
Individual anomalous payments	Addressed: ML score, anomaly percentile, amount-versus-average escalation, SHAP explanation.
Coordinated networks	Addressed for shared devices, flagged destinations, circular wallet transfers and linked clusters on synthetic data.
Unexplained decisions	Addressed: SHAP factors, policy notes and evidence-cited reports.
Money moving before review	Addressed: capture is gated and protected against bypass and duplication.
Real-world model accuracy	Not demonstrated. Models are trained and evaluated on synthetic data only.

2.4 Formal problem statement

> Given a PayPal order that a buyer has approved but that has not yet been captured, determine (a) whether the order is authentic and belongs to the requesting customer, (b) the risk that the payment is fraudulent, using the customer's history, the payment's context and the customer's relationships to other accounts, devices and destinations, and (c) whether the order may be captured automatically or must be held for an authenticated human decision, while producing an explanation and an auditable record for every step, and guaranteeing that a held order can never be captured without approval and that no order is captured twice.

PART 3

Project Objectives

3.1 Primary objectives

Objective	How it is met in the code	Why it matters
Detect suspicious transactions	Every transaction is scored by the primary ML model, the Isolation Forest and graph signals (`services/risk.py`).	Catches risky payments before money moves.
Estimate transaction risk	Fraud probability is mapped to LOW / MEDIUM / HIGH / CRITICAL by configurable thresholds, with escalation rules.	Lets the business choose how cautious to be.
Identify suspicious financial relationships	NetworkX graph over stored relationships detects shared devices, cycles, flagged destinations and suspicious clusters (`graph/engine.py`).	Organised fraud is invisible one payment at a time.
Investigate suspicious activity using AI	LangGraph workflow with seven tools; Groq LLM when configured, labelled rule-based fallback otherwise (`agent/`).	Saves analyst time and produces a consistent case file.
Integrate payment protection with PayPal Sandbox	Orders v2 create, get and capture; webhook verification; idempotency (`paypal/client.py`, `services/payments.py`).	Makes the risk decision consequential.
Provide explainable recommendations	SHAP top factors, policy notes, evidence keys E1…En in reports.	Reviewers and auditors can verify every claim.
Maintain a complete audit trail	SHA-256 hash-chained `audit_logs` table with a verification endpoint (`audit.py`).	Proves what happened and detects tampering.

3.2 Secondary objectives

- Real-time visibility: WebSocket event stream that refreshes the console.
- Role-based operation: ADMIN, REVIEWER, ANALYST and CUSTOMER roles.

- Reproducible demos: three scenarios that can be launched from the console.
- Honest evaluation: hold-out metrics only, primary model chosen by PR-AUC.
- Deployability: Docker Compose, Render Blueprint, Alembic migrations.

PART 4

Proposed Solution

4.1 The customer journey, step by step

1. **Customer selects a product.** The storefront lists 16 synthetic products from synthetic sellers. A cart can only contain items from one seller, because each checkout pays one merchant.
2. **Customer initiates PayPal Checkout.** The PayPal JavaScript SDK renders the PayPal button. Clicking it calls Sentinel's `POST /api/paypal/create-order` with the cart, a device identifier and an `Idempotency-Key`.
3. **Backend creates a PayPal order.** The server recalculates prices from its own catalogue, creates a Sentinel transaction with status `AWAITING_APPROVAL`, and calls PayPal `POST /v2/checkout/orders` with `intent = CAPTURE`. The PayPal order ID is returned to the browser.
4. **Customer approves in PayPal.** The buyer logs in to a PayPal Sandbox personal account and approves. No money has moved yet.
5. **Sentinel assesses risk before capture.** The browser calls `POST /api/paypal/capture-order`. The server fetches the order from PayPal, checks it is `APPROVED`, that amount, currency and reference match, and that the caller owns it.
6. **ML models analyse the transaction.** 26 point-in-time features are computed and scored by the primary model; the Isolation Forest gives an anomaly percentile; SHAP explains the score.
7. **Graph intelligence examines relationships.** The customer, device and destination are checked for shared-device rings, flagged destinations, circular loops and suspicious clusters.
8. **The AI agent investigates.** For HIGH and CRITICAL payments (by default) an investigation starts automatically in the background.
9. **A human reviewer examines the evidence.** The reviewer approves, rejects or escalates after re-entering their password and typing the transaction ID.
10. **Payment is captured or not.** LOW risk is captured automatically. An APPROVE decision triggers the capture. REJECT leaves the PayPal order uncaptured.
11. **Everything is recorded.** PayPal responses go to `payment_events`, decisions to `reviewer_decisions`, and every action to the hash-chained `audit_logs`.

4.2 Normal, medium-risk and high-risk scenarios

Scenario	Policy outcome	PayPal outcome	Human involvement
Normal (LOW)	`CLEARED`, then capture	Capture called with a stable `PayPal-Request-Id`; status `COMPLETED` comes from PayPal	None
Medium	`HELD_FOR_VERIFICATION`	Order stays `APPROVED`; nothing captured	Reviewer verifies and decides
High / Critical	`HELD_FOR_REVIEW`, AI investigation started	Order stays `APPROVED`; nothing captured	Reviewer decides; escalated cases need an ADMIN

> Important: a held payment is not a failed payment. The buyer has approved it in PayPal, but Sentinel has not asked PayPal to collect the money. The interface shows PayPal status and Sentinel status side by side so this is never confused.

PART 5

System Architecture

5.1 High-level architecture

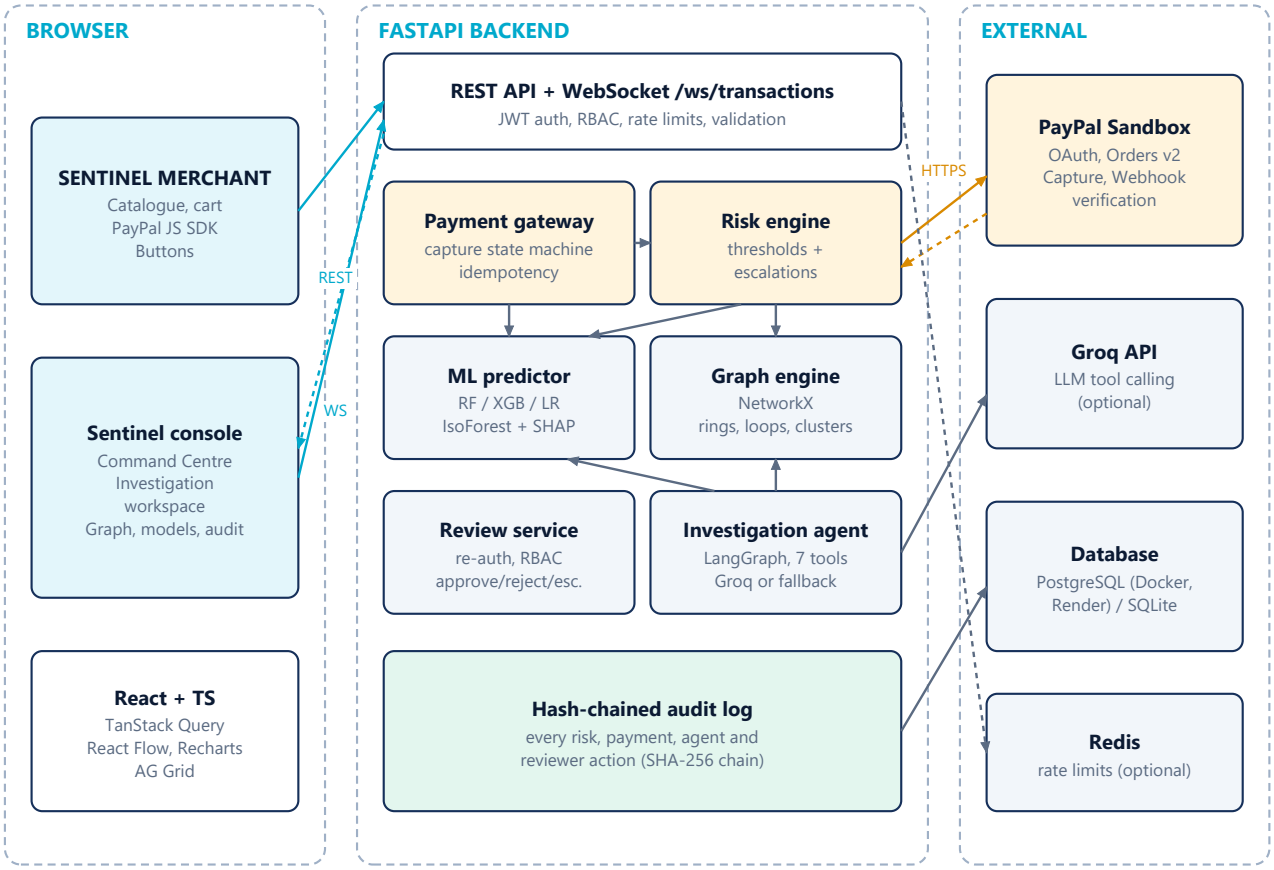


Figure 1: High-level system architecture

The system has three tiers. The **browser** runs a single React application with two areas: the SENTINEL MERCHANT storefront (shoppers) and the console (staff). The **FastAPI backend** exposes REST endpoints and one WebSocket, and contains the payment gateway, risk engine, ML predictor, graph engine, investigation agent and audit log. The **external services** are PayPal Sandbox (always used for real payments), Groq (optional, for the LLM agent), the database (PostgreSQL in Docker/Render, SQLite locally) and Redis (optional, for shared rate limiting).

Data flow in one sentence: the browser calls the API; the API stores state in the database, talks to PayPal for every payment operation, calls the risk engine (which uses the ML model and the graph), starts the agent for high-risk payments, records every step in the audit log, and pushes events back to the console over the WebSocket.

5.2 Frontend architecture

- **Routing** (`react-router-dom`): `/shop/*` for the storefront, `/console/*` for the console, `/login`.
- **State and data**: TanStack Query caches every API call. The `LiveProvider` opens one authenticated WebSocket and invalidates the relevant queries when events arrive, so every screen updates without manual refresh.
- **Auth**: `AuthProvider` stores the JWT in `sessionStorage` and exposes the current user and role. Console routes require a staff role.

- **Visualisation**: Recharts (dashboard and model charts), React Flow (relationship graph), AG Grid Community (transactions table), Framer Motion (transitions).
- **Payments**: `@paypal/react-paypal-js` renders PayPal Buttons using the public client ID returned by `GET /api/paypal/config`. The secret never reaches the browser.

5.3 Backend architecture

Module	Responsibility
`main.py`	FastAPI app, lifespan (migrations, first-run seeding and training, model artifact recovery), CORS, security headers, WebSocket.
`routers/`	`auth`, `transactions` (transactions, risk, reviews), `intel` (investigations, graph), `paypal` (checkout, webhook, storefront), `ops` (dashboard, audit, models, settings, demos).
`services/payments.py`	PayPal order lifecycle, capture state machine, webhook handling.
`services/risk.py`	Feature state, scoring, policy thresholds and escalations.
`services/review.py`	Reviewer decisions with re-authentication and RBAC.
`services/transactions.py`	Transaction creation and risk routing.
`ml/`	Feature engineering, training, SHAP explanation, prediction.
`graph/engine.py`	Graph persistence, NetworkX analysis, explorer views.
`agent/`	Seven investigation tools and the LangGraph workflow.
`audit.py`	Hash-chained audit log and chain verification.
`infra.py`	Rate limiter (Redis or memory) and WebSocket broadcaster.

5.4 Payment processing sequence

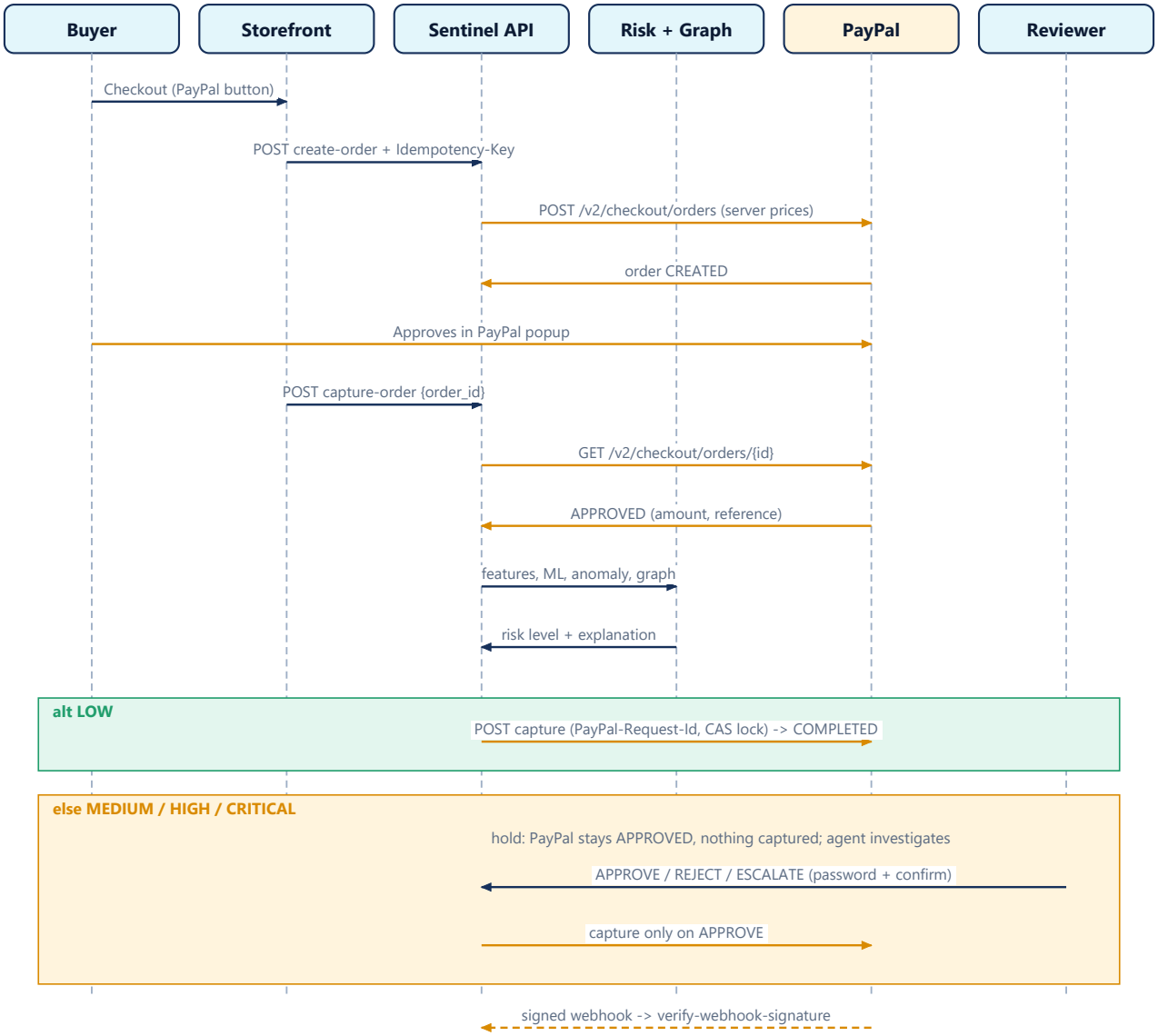


Figure 2: Payment processing sequence (Orders v2 with Sentinel risk gating)

The sequence shows the two independent approvals in the lifecycle: the buyer's approval in PayPal, and Sentinel's risk approval. The capture call to PayPal only happens if both exist. If the risk policy holds the payment, the reviewer's APPROVE decision replaces the automatic approval, and the same capture function is used, with the same protections.

5.5 Real-time communication

`ws://<host>/ws/transactions?token=<JWT>` accepts staff tokens only. The backend broadcasts `transaction.created`, `transaction.assessed`, `transaction.updated`, `payment.updated`, `investigation.step`, `investigation.completed` and `decision.made`. Because many events come from background threads (the agent, the simulator), the broadcaster uses `asyncio.run_coroutine_threadsafe` with the server's event loop.

5.6 Deployment architecture

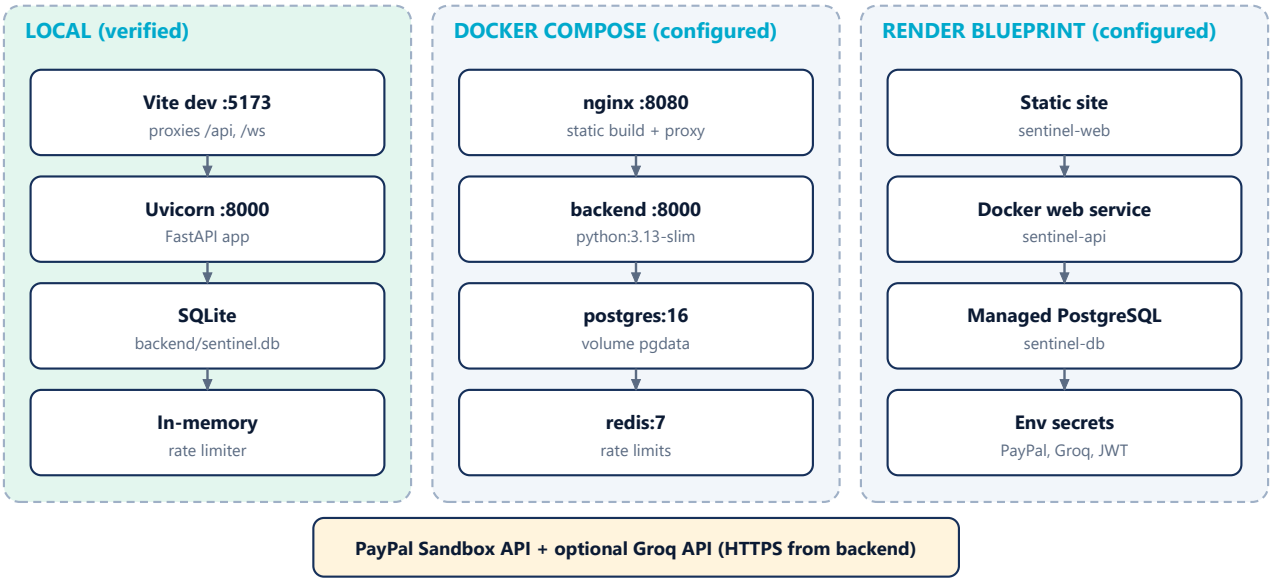


Figure 3: Deployment architecture

Three supported ways to run Sentinel:

Mode	Components	Status
Local	Uvicorn + Vite dev server, SQLite, in-memory rate limiting	Run and verified during development
Docker Compose	PostgreSQL 16, Redis 7, backend container, nginx serving the built frontend and proxying `/api` and `/ws`	Configuration written; not run during development because Docker was not running on the build machine
Render Blueprint	Managed PostgreSQL, Docker web service for the API, static site for the frontend	Configuration written; not deployed

On hosts with ephemeral disks, the backend retrains automatically at start-up if the active model file is missing.

PART 6

Technology Stack Explained

6.1 Frontend

Technology	What it is	Why chosen / problem solved	Where used	Talks to
React 18	UI component library	Component model for two applications in one bundle	All screens	TanStack Query, router
TypeScript 5.8	Typed JavaScript	Catches API shape mistakes at build time	Entire frontend	Compiler only
Vite 6	Build tool and dev server	Fast reloads; dev proxy for `/api` and `/ws`	Build and dev	Proxies to FastAPI
Tailwind CSS 3.4	Utility CSS framework	Consistent dark fintech design tokens	All styling	Build step
Framer Motion 12	Animation library	Smooth panel and list transitions	Cards, modals, feeds	React
Recharts 2	Chart library	Area, bar, line, pie, ROC/PR and SHAP charts	Command Centre, Fraud Intelligence, Model Analytics	API data via TanStack Query

Technology	What it is	Why chosen / problem solved	Where used	Talks to
React Flow (@xyflow/react 12)	Interactive node graph	Pan, zoom, hover and click on relationship graphs	Network Graph, Investigation workspace	`/api/graph/*`
AG Grid Community 33	Data grid	Filtering, sorting, quick search, pagination and CSV export over 2,000+ rows	Live Transactions	`/api/transactions`
TanStack Query 5	Server-state cache	Caching, polling, invalidation from WebSocket events	All data access	REST API
Lucide React	Icon set	Consistent icons	Everywhere	n/a
PayPal React SDK (@paypal/react-paypal-js 8)	Wrapper for PayPal JS SDK	Renders PayPal Buttons, handles approval popup	Checkout	PayPal + Sentinel API

AG Grid was evaluated as the brief asked: it adds real value for filtering and searching thousands of rows. Row grouping is an Enterprise feature, so risk "grouping" is done with filter chips instead.

6.2 Backend, data and infrastructure

Technology	What it is	Why chosen / problem solved	Where used
Python 3.13	Language	Same language for API, ML and graph code	Backend
FastAPI 0.115	Async web framework	Typed endpoints, automatic OpenAPI/Swagger, native WebSockets	`main.py`, `routers/`
Pydantic 2 / pydantic-settings	Validation and settings	Strict request validation; secrets loaded from environment	Request models, `config.py`
SQLAlchemy 2	ORM	One model layer for PostgreSQL and SQLite; atomic compare-and-set updates	`models.py`, services
Alembic 1.15	Migrations	Versioned schema changes	`alembic/versions/0001_initial_schema.py`
PostgreSQL 16	Relational database	Production database in Docker/Render	`docker-compose.yml`, `render.yaml`
SQLite	Embedded database	Zero-setup local development	Default `DATABASE_URL`
Redis 7 (optional)	In-memory store	Shared rate-limit counters across instances; falls back to memory if absent	`infra.py`
httpx	HTTP client	PayPal REST calls; injectable transport for tests	`paypal/client.py`
PyJWT, bcrypt	Auth libraries	Signed tokens, salted password hashing	`security.py`
Docker, Docker Compose	Containers	Reproducible multi-service environment	`Dockerfile`s, compose
Render	Cloud platform	Blueprint deployment of DB, API and static site	`render.yaml`
WebSockets (FastAPI/Starlette)	Push channel	Live console updates	`/ws/transactions`

6.3 Machine learning, graph and AI

Technology	Where used	Role
pandas 3, NumPy 2	`ml/train.py`, `ml/predictor.py`	Feature tables, metrics computation
scikit-learn 1.9	`ml/train.py`	Logistic Regression, Random Forest, Isolation Forest, imputation, scaling, metrics, stratified split

Technology	Where used	Role
XGBoost 3.4	`ml/train.py`	Gradient-boosted tree classifier
SHAP 0.54	`ml/explain.py`	TreeExplainer / LinearExplainer attributions per prediction and globally
joblib	`ml/train.py`	Model bundle persistence
NetworkX 3.7	`graph/engine.py`	Graph construction, connected components, `simple_cycles`, Kamada-Kawai layout
LangGraph 1.2	`agent/workflow.py`	State-machine orchestration of the investigation
Groq SDK 1.7	`agent/workflow.py`	LLM tool calling (default model `llama-3.3-70b-versatile`, configurable)

6.4 Payment

Component	Where used
PayPal JavaScript SDK (Buttons)	`frontend/src/pages/shop/Checkout.tsx`
OAuth 2.0 client credentials	`PayPalClient._access_token()`
Orders API v2: create, get, capture	`PayPalClient.create_order / get_order / capture_order`
Webhook signature verification API	`PayPalClient.verify_webhook()`
PayPal Sandbox	Default `PAYPAL_BASE_URL = https://api-m.sandbox.paypal.com`

PART 7

Machine Learning Deep Dive

7.1 Dataset structure

All data is **synthetic** and generated deterministically (seed 2026) by `backend/app/seed/synthetic.py`. No real people, cards or accounts are involved.

Item	Count	Notes
Customers	100	Home city and spending profile each; three are demo personas (Ava, Ben, Cara)
Merchants	25	5 in high-risk categories: gift cards, crypto on-ramp, electronics resale, gaming credits, prepaid mobile
Devices	150	Personal, secondary, shared household, attacker and emulator devices
Payment destinations	46	25 merchant payout accounts, 19 customer-owned wallets, 2 flagged "mule" wallets
Products	16	Storefront catalogue
Historical transactions	2,000	Spread over 90 days
Labelled fraud	142 (7.1%)	Ground-truth label `is_fraud` exists only for synthetic history

The generator creates labelled behaviour patterns, including deliberately hard cases:

Pattern (scenario tag)	Transactions	Label
NORMAL	1,800	Legitimate (3% random declines)
RING_DEVICE_FARM	60	Mostly fraud (92%): 9 accounts, shared emulator devices, night bursts, mule wallets
CIRCULAR_RING	24	Fraud: 4 accounts passing money in a loop
ACCOUNT_TAKEOVER	20	Fraud: new device, far-away location, 8 to 25 times normal spend

Pattern (scenario tag)	Transactions	Label
CARD_TESTING	26	Fraud: tiny failed attempts then a large purchase
FRIENDLY_FRAUD	16	Fraud that looks completely normal (label noise)
LEGIT_LARGE	22	Legitimate large purchases (hard negatives)
P2P_LEGIT	32	Legitimate wallet transfers

Friendly fraud and legitimate large purchases are included on purpose: they stop the models from reaching unrealistic perfect scores.

7.2 Pipeline overview

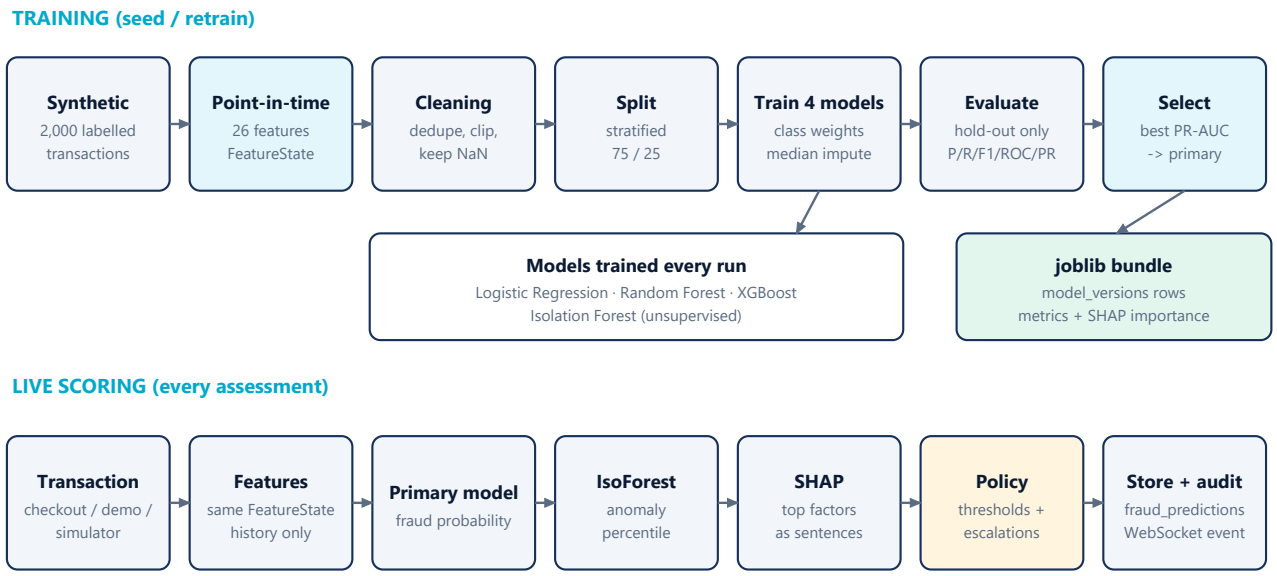


Figure 4: Machine-learning fraud detection pipeline

7.3 Feature engineering (point-in-time)

A transaction's features are computed **only from transactions that happened before it**. `FeatureState` replays history in time order for training, and the same class is used for live scoring. This prevents "looking into the future", and it guarantees the model sees identical feature definitions in training and production (no train/serve skew).

Group	Features (26 total)
Amount	`amount`, `log_amount`
Frequency	`txn_count_1h`, `txn_count_24h`, `txn_count_7d`, `hours_since_prev`
Customer baseline	`cust_avg_amount`, `amount_to_avg_ratio`, `amount_zscore`
Merchant	`merchant_risk_score`, `merchant_high_risk_category`, `is_p2p_transfer`
Device	`device_distinct_customers`, `device_new_for_customer`, `customer_distinct_devices_30d`
Geography	`geo_distance_prev_km`, `geo_velocity_kmh`, `distance_from_home_km` (haversine)
Failures	`failed_count_24h`, `failed_count_7d`
Network	`destination_distinct_customers`, `destination_flagged`, `customer_flagged_dest_interactions`
Time	`hour_of_day`, `is_night`, `is_first_txn`

7.4 Preprocessing, split and class imbalance

- **Cleaning**: drop unlabelled rows, duplicate IDs and non-positive amounts; clip extreme values (speed at 5,000 km/h, ratio at 200, z-score to [-50, 200]).
- **Missing values**: the first transaction of a customer has no history, so some features are missing. They stay `NaN` and are filled with the training median inside each model pipeline (`SimpleImputer`), so the test set never influences imputation.
- **Split**: stratified 75/25 hold-out with `random_state=42`: 1,500 training rows and 500 test rows (36 fraud cases in the test set).
- **Class imbalance** (about 1 fraud per 13 legitimate): `class_weight="balanced"` for Logistic Regression, `class_weight="balanced_subsample"` for Random Forest, and `scale_pos_weight = negatives / positives` for XGBoost.

7.5 The four algorithms

Logistic Regression

In plain words: it gives each feature a weight, adds them up, and squashes the total into a probability between 0 and 1.

Technically: p(fraud | x) = 1 / (1 + e^-(w·x + b)). The weights are learned by minimising the log-loss with L2 regularisation (C = 0.5). Features are standardised first.

Advantages: fast, stable, easy to interpret. **Limitations:** it only draws straight decision boundaries, so interactions (for example "new device AND far away") must be engineered by hand. **Role in Sentinel:** a transparent baseline. In v1 it had the best ROC-AUC (0.964) but lower precision (0.842).

Random Forest

In plain words: it grows hundreds of decision trees, each on a random sample of the data and features, and lets them vote.

Technically: bagging of CART trees (300 trees, min_samples_leaf = 2). The predicted probability is the average of the trees' leaf class proportions. Randomisation reduces variance compared with a single tree.

Advantages: captures non-linear interactions, robust to outliers, needs little tuning. **Limitations:** larger model, probabilities are not calibrated, less extrapolation. **Role in Sentinel:** the **primary model in v1**, because it had the highest PR-AUC (0.914).

XGBoost

In plain words: it builds trees one after another, each new tree fixing the mistakes of the previous ones.

Technically: gradient boosting. Each tree is fitted to the gradient of the log-loss; leaf weights use second-order (Hessian) information; regularisation (`reg_lambda = 1.5`, depth 4, learning rate 0.07, 250 trees, row and column subsampling) controls overfitting. `scale_pos_weight` up-weights fraud cases.

Advantages: usually the strongest family on tabular data; handles missing values; fast. **Limitations:** more hyperparameters; can overfit small data. **Role in Sentinel:** trained and evaluated every time; it would be selected automatically if it had the best PR-AUC. In v1 it came third by PR-AUC (0.903), so it was not the primary model. This is reported honestly rather than forcing XGBoost to be primary.

Isolation Forest

In plain words: unusual points are easier to "isolate" with random cuts than normal ones.

Technically: an ensemble of random isolation trees; the anomaly score is based on the average path length needed to isolate a point (shorter path = more anomalous). It is trained **without labels**. Sentinel converts the raw

score into a percentile of the training scores ("more anomalous than 95% of training transactions").

Advantages: detects new kinds of unusual behaviour that labels do not cover. **Limitations:** "unusual" is not the same as "fraudulent". **Role in Sentinel:** a second opinion used by the policy (anomaly percentile at or above 0.98 raises LOW to at least MEDIUM) and shown to reviewers.

7.6 Model selection and evaluation

The primary model is chosen automatically as the supervised model with the highest **PR-AUC** on the hold-out set. All numbers below are the stored results of model version v1 (trained on 1,500 rows, evaluated on 500 rows the models never saw), at threshold 0.5 for the supervised models. Data generation and training are deterministic, so a fresh seed reproduces exactly these numbers:

Model	Precision	Recall	F1	ROC-AUC	PR-AUC	TN	FP	FN	TP
Random Forest (primary)	0.970	0.889	0.927	0.954	0.914	463	1	4	32
Logistic Regression	0.842	0.889	0.865	0.964	0.911	458	6	4	32
XGBoost	0.939	0.861	0.899	0.958	0.903	462	2	5	31
Isolation Forest (unsupervised)	0.849	0.778	0.812	0.944	0.862	459	5	8	28

> These metrics describe performance on synthetic data generated by the project itself. They demonstrate that the pipeline is correct and evaluated honestly; they are not evidence of real-world fraud detection accuracy.

7.7 Understanding the metrics

Metric	Meaning	In fraud terms
Precision	TP / (TP + FP)	Of the payments we flagged, how many were really fraud. Low precision = annoyed genuine customers.
Recall	TP / (TP + FN)	Of all fraud, how much we caught. Low recall = fraud slips through.
F1	Harmonic mean of precision and recall	One number that balances both.
ROC-AUC	Probability a random fraud case is scored above a random legitimate one	Threshold-free ranking quality; can look optimistic when fraud is rare.
PR-AUC (average precision)	Area under the precision-recall curve	Focuses on the rare fraud class; the most informative single number under imbalance.
Confusion matrix	TN, FP, FN, TP counts	Shows exactly which mistakes are made.
False positive	Genuine payment flagged	Customer friction, manual review cost.
False negative	Fraud not flagged	Financial loss, chargebacks.

Why accuracy is not enough: with 7% fraud, a model that says "never fraud" is 93% accurate and catches nothing. That is why Sentinel reports precision, recall, F1, ROC-AUC and PR-AUC and selects by PR-AUC.

7.8 Explainable AI with SHAP

SHAP (SHapley Additive exPlanations) splits one prediction into contributions from each feature, based on Shapley values from cooperative game theory: each feature's contribution is its average marginal effect over all orderings of features. For each prediction, the contributions plus the model's baseline add up to the model output.

Sentinel uses `TreeExplainer` for tree models and `LinearExplainer` for Logistic Regression. The explanation space is reported with every prediction: Random Forest values are in probability space; XGBoost and Logistic Regression values are in log-odds.

For each payment, positive contributions are converted into sentences a reviewer can check, for example "*Amount is 43.4x the customer's historical average*" or "*Device has been used by 4 distinct accounts*". Globally, the mean absolute SHAP value on the test set ranks the features. The top ten for v1 are:

Rank	Feature	Mean abs SHAP
1	amount	0.0701
2	log_amount	0.0698
3	distance_from_home_km	0.0662
4	hour_of_day	0.0636
5	is_night	0.0382
6	hours_since_prev	0.0311
7	destination_distinct_customers	0.0211
8	geo_distance_prev_km	0.0204
9	merchant_risk_score	0.0182
10	geo_velocity_kmh	0.0176

Note: `hour_of_day` and `is_night` rank highly because the synthetic fraud ring operates at night. That is a property of the synthetic generator and a known limitation (Part 17).

7.9 Risk scoring workflow

1. Compute features from history (`FeatureState.features`).
2. Primary model gives `fraud_probability`; Isolation Forest gives `anomaly_score` (percentile); SHAP gives contributions.
3. Classify: LOW below 0.30, MEDIUM below 0.70, HIGH below 0.90, otherwise CRITICAL (editable by ADMIN).
4. Escalations (they can only raise risk): anomaly percentile at least 0.98 gives at least MEDIUM; amount at least 10 times average gives at least MEDIUM; graph signals (flagged destination, device shared by 3 or more accounts, circular loop) give at least HIGH.
5. Store a `fraud_predictions` row with the probability, scores, level, action, top factors, SHAP values, features, policy notes and latency; write an audit entry; broadcast an event.

Recommended actions: LOW `AUTO_APPROVE`, MEDIUM `ADDITIONAL_VERIFICATION`, HIGH `MANUAL_REVIEW`, CRITICAL `MANUAL_REVIEW_AND_INVESTIGATE`.

7.10 Retraining and feedback

`POST /api/models/retrain` (ADMIN) creates a new model version. It uses ground-truth labels for synthetic history and reviewer decisions for live payments (REJECT = fraud, APPROVE = legitimate). Previous versions remain stored.

PART 8

Graph Intelligence

8.1 Why transaction-level analysis is not enough

A model scores one payment using that customer's history. A fraud ring is designed so that each payment looks normal. The evidence is in the **connections**: one device used by several accounts, several accounts paying the same wallet, money going round in a circle. Graph analysis makes those connections visible.

8.2 Architecture

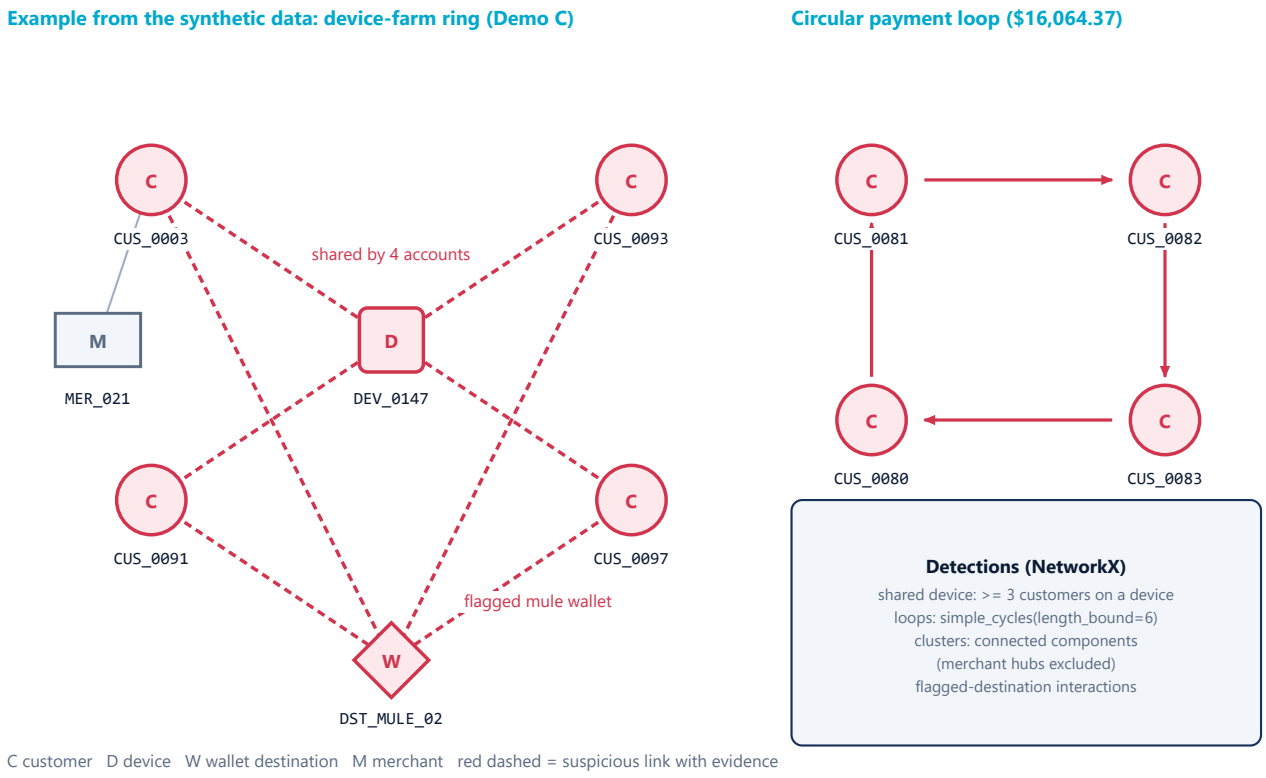


Figure 5: Graph intelligence: suspicious structures found in the synthetic data

Every transaction writes relationships into two tables: `graph_entities` (nodes) and `graph_relationships` (edges, with a weight counting repeats). The engine loads them into a NetworkX directed graph and then builds an **entity projection** that collapses transaction nodes into aggregated edges between customers, devices, destinations and merchants, each carrying a count, total amount and transaction IDs.

Node type	Example
CUSTOMER	CUS_0003 (Cara Novak)
MERCHANT	MER_021 (VaultGift Digital)
DEVICE	DEV_0147 (Android emulator)
DESTINATION	DST_MULE_02 (flagged external wallet)
TRANSACTION	TXN_100542

Edge type	Meaning
USES_DEVICE	customer used device

Edge type	Meaning
MADE_PAYMENT	customer made transaction
VIA_DEVICE	transaction made on device
PAID_TO	transaction paid destination
PAID_MERCHANT	transaction paid merchant
OWNS_WALLET	customer owns wallet destination

8.3 Detection algorithms

- **Shared devices:** device nodes connected to 3 or more customers.
- **Circular payments:** a directed customer-to-customer money-flow graph (payer to the owner of the wallet paid) searched with `networkx.simple_cycles(length_bound=6)`; loops of 3 or more accounts are reported.
- **Suspicious clusters:** connected components over customers, devices and wallets. Merchants and merchant payout accounts are excluded, otherwise everyone would be connected through popular shops. A cluster is suspicious if it contains a shared device, a flagged destination, a cycle member, or has mean account risk of at least 0.5.
- **Flagged destination interactions:** every account and payment that reached a flagged wallet.
- **Graph risk score:** a 0 to 1 score per customer from these signals (cycle 0.9, flagged destination 0.85, shared device 0.6 plus 0.05 per account up to 8, suspicious cluster 0.5 plus half the mean risk).
- **Traversal for views:** two-hop breadth-first expansion that never passes through merchant hubs; layout by Kamada-Kawai per connected component.

Community detection (for example Louvain) is **not implemented**; connected components are used. See Future Scope.

8.4 What the graph found in the synthetic data

Finding	Detail
Entities / relationships	2,321 / 8,097 (freshly seeded database)
Shared device DEV_0147	4 accounts: CUS_0003, CUS_0091, CUS_0093, CUS_0097
Shared device DEV_0148	3 accounts: CUS_0090, CUS_0094, CUS_0096
Circular loop	CUS_0081 to CUS_0082 to CUS_0083 to CUS_0080 and back, $16,064.37 in total
Flagged DST_MULE_01	16 payments, $11,238.59, from 7 accounts
Flagged DST_MULE_02	18 payments, $9,831.23, from 9 accounts
Suspicious clusters	4 (cluster 1: 9 accounts, 2 shared devices, 2 flagged destinations; cluster 8: the 4-account loop)

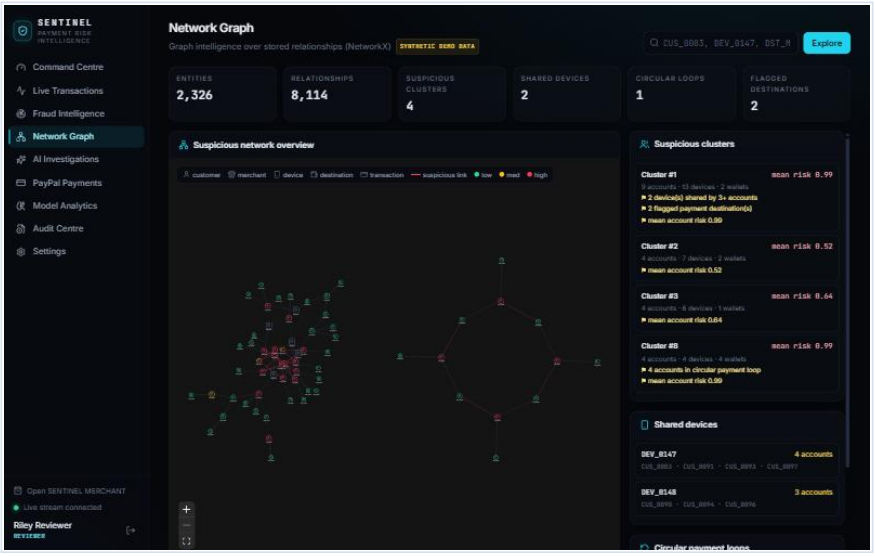


Figure 6: Network Graph overview: the device-farm cluster (left) and the four-account circular payment loop (right).

8.5 How graph findings become evidence

The `GraphInvestigationTool` returns the customer's signals; the `EvidenceCollectionTool` converts each into a typed evidence item such as "*E6 HIGH SHARED_DEVICE: Device DEV_0147 is shared by 4 accounts (CUS_0003, CUS_0091, CUS_0093, CUS_0097)*". Explorer edges carry the same evidence strings and transaction IDs, so a reviewer can click a red edge and open the underlying transactions.

PART 9

Agentic AI Investigation System

9.1 What "agentic" means here

Type	Behaviour
Traditional chatbot	Answers questions from its training; no access to your data.
Basic LLM application	Sends a prompt (perhaps with some data) and returns text.
Tool-using agent	The model decides to call functions (tools) that fetch real data, then uses the results.
Multi-step investigation workflow	A defined sequence of steps with shared state, tool calls, validation and a structured output.

Sentinel's investigator is the last kind: a LangGraph state machine whose fixed steps always run the core tools, followed by an analysis step in which an LLM may call more tools before submitting a structured report.

9.2 The workflow

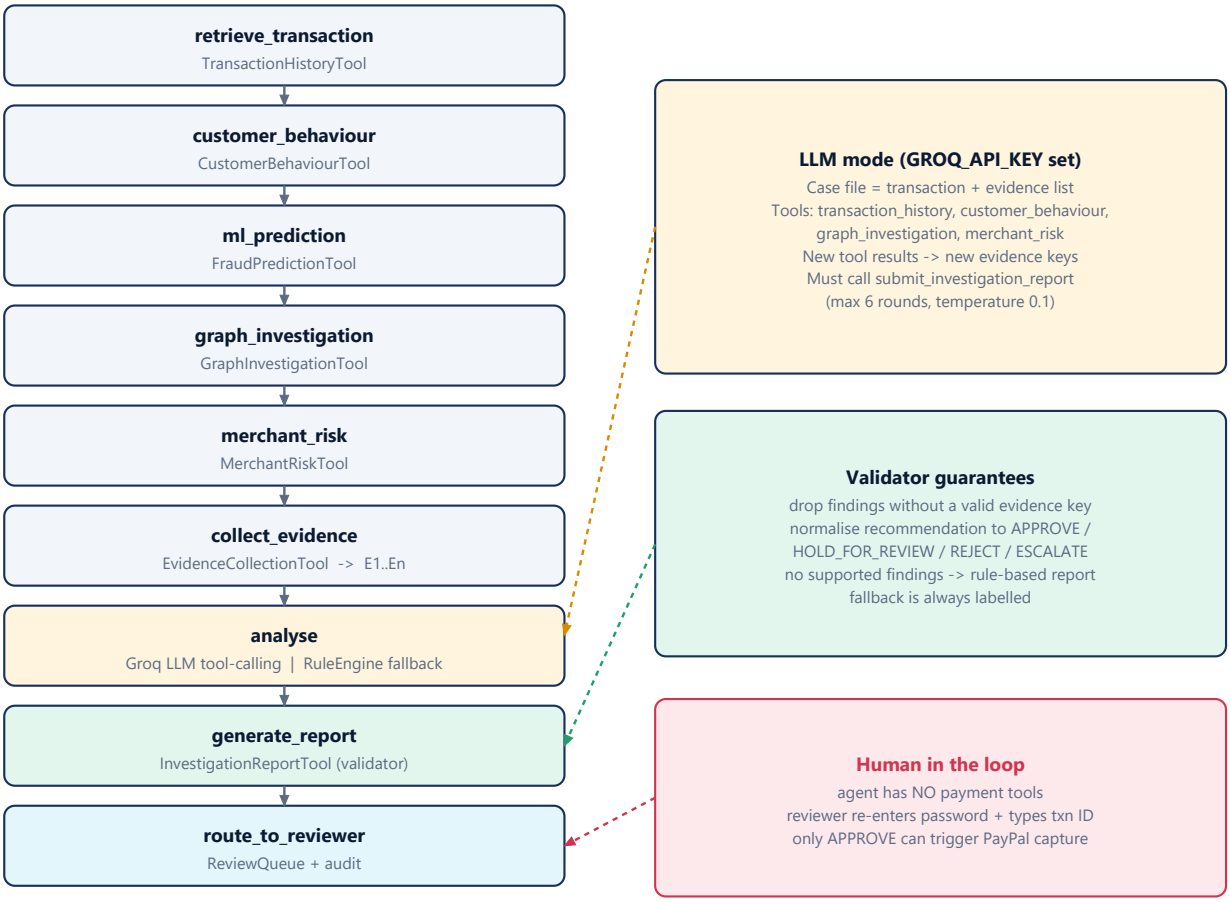


Figure 7: LangGraph investigation agent workflow

Node	Tool	Output
retrieve_transaction	TransactionHistoryTool	Focal transaction and up to 25 prior transactions
customer_behaviour	CustomerBehaviourTool	Average, median, devices, failures, distance from home, speed since previous
ml_prediction	FraudPredictionTool	Stored prediction, SHAP factors, policy notes (assesses first if none exists)
graph_investigation	GraphInvestigationTool	Shared devices, flagged destinations, cycles, cluster
merchant_risk	MerchantRiskTool	Merchant profile, or destination analysis for peer-to-peer transfers
collect_evidence	EvidenceCollectionTool	Evidence items E1…En with severity and "mitigating" flags
analyse	Groq LLM, or RuleEngine	Draft report
generate_report	InvestigationReportTool	Validated report; evidence stored in `investigation_evidence`
route_to_reviewer	ReviewQueue	Recommendation routed; audit entry

State management: a typed `InvState` dictionary carries the transaction ID, related entity IDs, tool outputs, draft, report and mode between nodes. Each step's start time, duration, status and summary is saved to `investigations.steps` and broadcast over the WebSocket, so the console shows progress live.

9.3 Tool calling with Groq

When `GROQ_API_KEY` is set, the analysis step sends the case file (transaction plus evidence list) to Groq with five tool definitions: `transaction_history`, `customer_behaviour`, `graph_investigation`, `merchant_risk` and `submit_investigation_report`. The model may call the first four to look at related entities (for example another account on the shared device). Their results become new evidence keys. It must finish by calling `submit_investigation_report` with a structured payload: `risk_level`, `summary`, `findings[{statement, evidence_keys}]`, `recommendation` (APPROVE, HOLD_FOR_REVIEW, REJECT or ESCALATE) and `rationale`. The loop is limited to 6 rounds, with temperature 0.1.

> Verification status: the Groq tool-calling path is implemented but was **not exercised** during development because no Groq API key was available. All investigations shown in this document used the deterministic fallback, and are labelled as such.

9.4 Hallucination prevention

1. Evidence is produced deterministically from tool outputs, never by the LLM.
2. The LLM only sees the case file and tool results.
3. `InvestigationReportTool` removes every finding that cites no valid evidence key and lists it under `rejected_findings`.
4. An invalid recommendation is replaced with `HOLD_FOR_REVIEW`.
5. If no supported findings remain, the rule-based report is used instead, with the reason recorded.
6. Tests check that invented keys (for example `E999`) and uncited claims are removed.

9.5 Error handling and fallback

If Groq is not configured, or the call fails, times out or does not submit a report within the step budget, the agent produces a rule-based report marked `DETERMINISTIC_FALLBACK` with the reason (for example "*GROQ_API_KEY not set*"). The UI shows "Deterministic rule-based report: not LLM-generated". A tool error marks that step as `ERROR` and the workflow continues; an unexpected exception marks the investigation `FAILED` with the error message.

9.6 Why the agent recommends but never acts

Capture, refund and rejection are irreversible financial actions, and LLMs can be wrong. The agent has **no payment tools**. Its only output is a recommendation placed in the review queue. A human with the REVIEWER or ADMIN role must decide, after re-entering their password and typing the transaction ID.

9.7 End-to-end example (actual run, Demo C)

Step	Duration	Result
Retrieve transaction	15.4 ms	CUS_0003 has 8 transactions; focal TXN_563TG7BVLJ $500.00
Analyse customer behaviour	6.1 ms	7 prior transactions, average $327.57, 2 devices, 0 failures
ML prediction + SHAP	4.3 ms	random_forest v1 scored 0.333 (final level HIGH); anomaly percentile 0.947
Graph connections	3.0 ms	1 shared device, 1 flagged destination link, suspicious cluster #1 (9 accounts)
Merchant risk	5.7 ms	VaultGift Digital: risk 0.72, chargeback rate 6.1%
Collect evidence	1.8 ms	8 evidence items, 4 high severity
Analysis	1.5 ms	Deterministic (Groq not configured)
Generate report	9.5 ms	REJECT, 5 evidence-backed findings

Step	Duration	Result
Route to reviewer	5.7 ms	Recommendation sent to reviewer queue

The graph step reads the cached graph engine, which is updated incrementally, so it takes milliseconds. The reviewer then rejected the payment with the reason "*Shared emulator device with 4 ring accounts and flagged mule destination (E5, E6).*"

PART 10

PayPal Integration

10.1 Developer platform setup

Item	How Sentinel uses it
PayPal Developer Dashboard	A Sandbox REST app is created under Apps & Credentials.
Sandbox environment	`PAYPAL_BASE_URL=https://api-m.sandbox.paypal.com`; no real money moves.
Sandbox business account	Owns the REST app and receives captured funds.
Sandbox personal account	Used by the tester to approve payments in the PayPal popup.
Client ID	Public; sent to the browser by `GET /api/paypal/config` to load the JS SDK.
Client secret	Server-only environment variable; never sent to the browser or logged.
Webhook ID	`PAYPAL_WEBHOOK_ID`, required to verify webhook signatures.

10.2 Calls Sentinel makes

Step	PayPal API	Sentinel code
OAuth token	`POST /v1/oauth2/token` (client credentials, cached until expiry)	`_access_token()`
Create order	`POST /v2/checkout/orders` with `intent=CAPTURE`, item breakdown, `reference_id`/`custom_id` = Sentinel transaction ID, `PayPal-Request-Id`	`create_order()`
Verify approved order	`GET /v2/checkout/orders/{id}`	`get_order()`
Capture	`POST /v2/checkout/orders/{id}/capture` with `PayPal-Request-Id: capture-{order_id}`	`capture_order()`
Verify webhook	`POST /v1/notifications/verify-webhook-signature`	`verify_webhook()`

10.3 Payment lifecycle, step by step

1. **PayPal order creation.** Sentinel creates the order; PayPal returns `status: CREATED`. Sentinel stores `paypal_status=CREATED`, `sentinel_status=AWAITING_BUYER_APPROVAL`.
2. **Customer approval.** The buyer approves in PayPal. PayPal's order becomes `APPROVED`. No funds have moved.
3. **Sentinel risk assessment.** On `capture-order`, Sentinel GETs the order, requires `APPROVED`, and checks amount, currency, reference and ownership. A mismatch sets `INTEGRITY_FAILED` and returns HTTP 409. Then it scores the risk.
4. **PayPal capture.** Only for LOW risk (automatic) or after a reviewer's APPROVE.
5. **Confirmed payment status.** Capture ID and capture status are copied from PayPal's response; `PAYMENT.CAPTURE.*` webhooks (when configured) update the same fields.

10.4 Two status fields, never mixed

Field	Written from	Example values
`paypal_status`, `capture_status`, `capture_id`	PayPal API responses and verified webhooks only	CREATED, APPROVED, COMPLETED; capture COMPLETED / PENDING / DENIED
`sentinel_status`	Sentinel's policy and reviewers	AWAITING_BUYER_APPROVAL, CLEARED, HELD_FOR_VERIFICATION, HELD_FOR_REVIEW, ESCALATED, APPROVED_AUTO, APPROVED_BY_REVIEWER, REJECTED, CAPTURE_FAILED, INTEGRITY_FAILED

A held payment therefore appears as PayPal `APPROVED`, capture "not captured", Sentinel `HELD_FOR_REVIEW`. The shopper sees: "*Payment approved by buyer but held by Sentinel for human review. Funds have NOT been captured.*"

10.5 Idempotency and duplicate-capture prevention

- **Order creation:** the client sends an `Idempotency-Key`; it is stored as a unique column and forwarded as `PayPal-Request-Id`. A repeated request returns the existing order.
- **Capture lock:** `capture_state` moves from NONE or FAILED to IN_PROGRESS with an atomic `UPDATE ... WHERE capture_state IN ('NONE','FAILED')`. Only the request that changes the row calls PayPal; any other returns "already performed or in progress".
- **PayPal-side idempotency:** a stable `PayPal-Request-Id = capture-{order_id}` means retries cannot capture twice at PayPal either.
- **Reconciliation:** if PayPal answers `ORDER_ALREADY_CAPTURED`, Sentinel GETs the order and copies the real capture.
- **Authorisation re-check:** the capture function itself requires a LOW assessment (automatic) or a stored APPROVE decision (reviewer).

10.6 Webhooks

Webhooks are verified with PayPal's verification API using `PAYPAL_WEBHOOK_ID`. Unverified events return HTTP 400 and are audited as `WEBHOOK_REJECTED`; duplicates (same event ID) are ignored. Handled events: `CHECKOUT.ORDER.APPROVED`, `PAYMENT.CAPTURE.COMPLETED`, `.DENIED`, `.PENDING`, `.REFUNDED`, `.REVERSED`. Webhooks need a public HTTPS URL, so they were not received during local development.

10.7 Error handling

PayPal errors raise `PayPalError` with HTTP status, issue code and `paypal-debug-id`. The error body is stored in `payment_events` and returned to the client as HTTP 502 with the debug ID. If credentials are missing, payment endpoints return HTTP 503 and the checkout screen explains how to configure PayPal; Sentinel never shows a simulated success.

10.8 Honest scope and verification status

- Sentinel **does not** access PayPal's internal fraud systems. It is a merchant-side risk layer that uses public PayPal REST APIs.
- The integration was tested end to end against a **mock PayPal transport** (a test double emulating Sandbox semantics, used only in tests).
- **Live Sandbox calls were not executed during development**, because Sandbox credentials were not available on the build machine. Configuring `PAYPAL_CLIENT_ID` and `PAYPAL_CLIENT_SECRET` and running Demo A and Demo C through checkout is the remaining verification step.

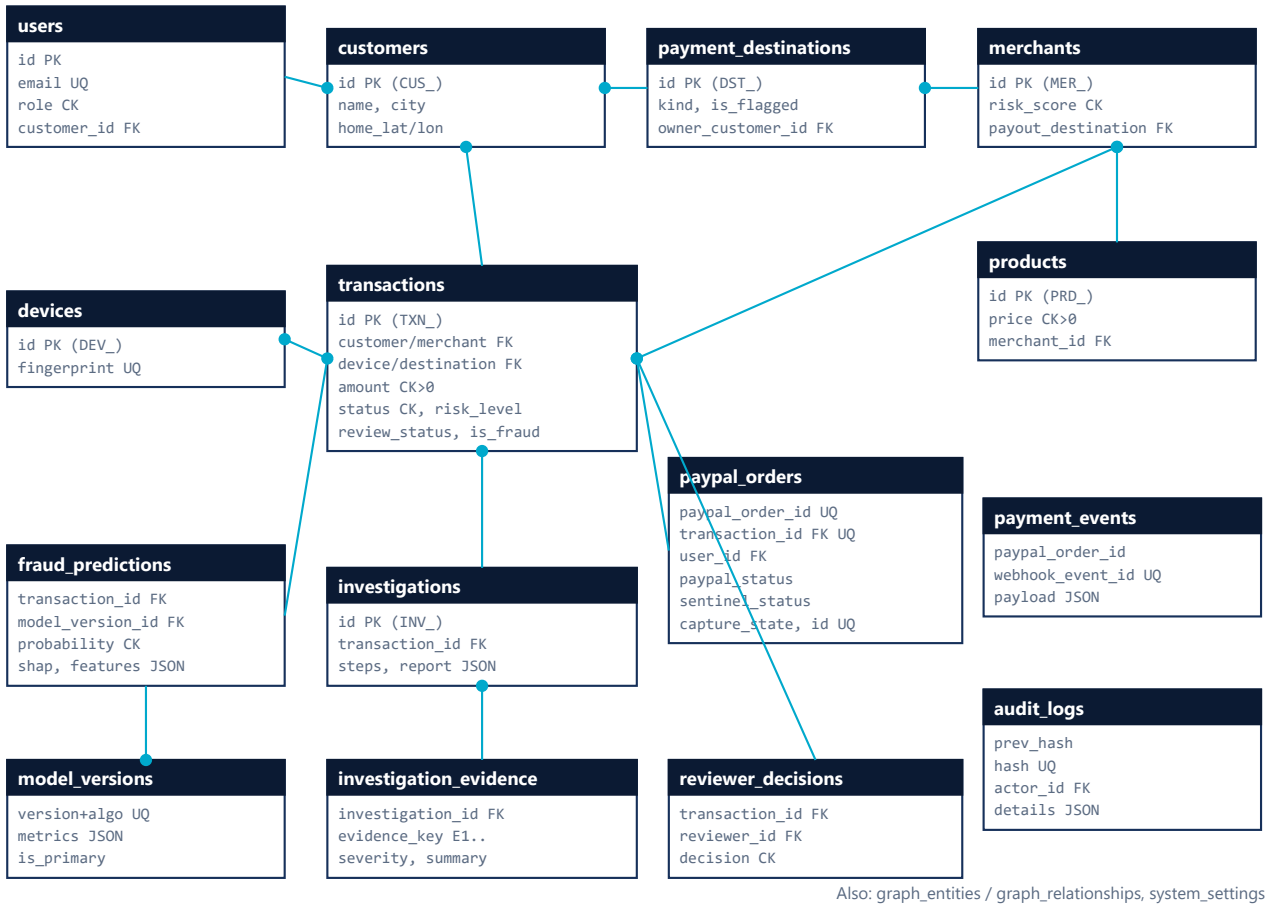
10.9 Why PayPal is central, not an add-on

The project's core guarantee, "risky money does not move without a human", only means something if Sentinel controls a real capture step. PayPal's order and capture separation is what makes risk gating possible, and PayPal's request IDs and webhook signatures are part of the safety design.

PART 11

Database and API Documentation

11.1 Entity relationships



Also: graph_entities / graph_relationships, system_settings

Figure 8: Database relationships (main tables and foreign keys)

11.2 Tables

All tables are created by Alembic migration `0001_initial_schema`. Timestamps are timezone-aware.

Table	Purpose	Key fields	Foreign keys	Constraints
users	Staff and shopper accounts	id PK, email (unique), password_hash, role, is_active	customer_id to customers	role in ADMIN, REVIEWER, ANALYST, CUSTOMER
customers	Synthetic customers	id PK (CUS_...), name, city, country, home_lat/lon, segment	none	none

Table	Purpose	Key fields	Foreign keys	Constraints
merchants	Sellers	id PK (MER_...), category, risk_score, high_risk_category, chargeback_rate	payout_destination_id to payment_destinations	0 <= risk_score <= 1
devices	Device fingerprints	id PK (DEV_...), fingerprint (unique), type, os, is_flagged	none	unique fingerprint
payment_destinations	Where money lands	id PK (DST_...), kind, is_flagged, flag_reason	owner_customer_id to customers	none
products	Storefront catalogue	id PK (PRD_...), name, price, stock	merchant_id to merchants	price > 0
transactions	Every payment	id PK (TXN_...), amount, lat/lon, channel, status, risk_level, review_status, is_fraud	customer, merchant, device, destination	amount > 0; status in allowed list; indexes on customer+time, device+time
fraud_predictions	Every risk assessment	probability, anomaly_score, graph_risk_score, risk_level, action, top factors, SHAP, features, policy notes, latency	transaction_id, model_version_id	0 <= probability <= 1
model_versions	Trained models	version, algorithm, is_active, is_primary, metrics JSON, artifact_path	none	unique (version, algorithm)
investigations	Agent runs	id PK (INV_...), status, mode, steps, report, recommendation	transaction_id, started_by_id	none
investigation_evidence	Evidence items	evidence_key, tool_name, category, severity, summary, data	investigation_id	none
graph_entities	Graph nodes	entity_type, ref_id, label, risk_score, flagged	none	unique (entity_type, ref_id)
graph_relationships	Graph edges	rel_type, weight	source_id, target_id to graph_entities	unique (source, target, type)
paypal_orders	PayPal order mirror	paypal_order_id (unique), amount, items, create_idempotency_key (unique), paypal_status, sentinel_status, capture_state, capture_id (unique), capture_status	transaction_id (unique), user_id	amount > 0
payment_events	PayPal API responses and webhooks	source, event_type, webhook_event_id (unique), signature_verified, http_status, paypal_debug_id, payload	none (indexed paypal_order_id)	unique webhook_event_id
reviewer_decisions	Human decisions	decision, reason, risk_level_at_decision, payment_outcome	transaction, investigation, reviewer	decision in APPROVE, REJECT, ESCALATE
audit_logs	Hash-chained audit trail	actor_label, action, entity, details, prev_hash, hash (unique)	actor_id to users	unique hash
system_settings	Key-value settings (risk policy)	key PK, value JSON	none	none

No card numbers, bank details or PayPal credentials are stored. Payer email and payer ID are stored only if PayPal returns them.

11.3 API endpoints

Authentication: `Authorization: Bearer <JWT>` from `POST /api/auth/login`. "Staff" means ADMIN, REVIEWER or ANALYST. Interactive documentation is at `/docs` (Swagger) and `/redoc`.

Method	Route	Purpose	Auth
POST	/api/auth/login	Email + password to JWT (rate limit 10/min)	Public
GET	/api/auth/me	Current user	Any user
POST	/api/transactions	Create a synthetic transaction and assess it	Staff
GET	/api/transactions	List with filters: risk_level, status, channel, review_status, customer_id, q, limit, offset	Staff
GET	/api/transactions/{id}	Full case: customer, merchant, device, destination, predictions, investigations, decisions, PayPal order, events, audit	Staff
POST	/api/risk/assess	Re-assess a transaction with the active model	Staff
GET	/api/risk/metrics	Active model metrics and risk distribution	Staff
POST	/api/investigations/start	Start the agent (asynchronous)	Staff
GET	/api/investigations	List investigations	Staff
GET	/api/investigations/{id}	Steps, report, evidence	Staff
GET	/api/graph/overview	Suspicious clusters, shared devices, cycles, flagged destinations and graph	Staff
GET	/api/graph/entity/{id}	Neighbourhood of an entity (hops 1-3)	Staff
GET	/api/graph/transaction/{id}	Neighbourhood of a transaction	Staff
GET	/api/paypal/config	Public client ID and status (no secrets)	Public
POST	/api/paypal/create-order	Create PayPal order; header `Idempotency-Key` required	Shopper
POST	/api/paypal/capture-order	Verify, assess, then capture or hold	Order owner
GET	/api/paypal/order/{id}	Refresh state from PayPal	Owner or staff
GET	/api/paypal/orders	Orders (own for shoppers, all for staff)	Any user
GET	/api/paypal/events	PayPal API responses and webhooks	Staff
POST	/api/paypal/webhook	Verified PayPal webhooks (rate limit 300/min)	PayPal signature
POST	/api/reviews/{txn}/decision	APPROVE / REJECT / ESCALATE with password and confirmation	Reviewer, Admin
GET	/api/reviews/queue	Payments awaiting a decision	Staff
GET	/api/dashboard/metrics	KPIs, time series, distributions, network and model summary	Staff
GET	/api/audit	Audit log with filters	Staff
GET	/api/audit/verify	Verify the hash chain	Staff
GET	/api/models	All model versions and metrics	Staff
GET / POST	/api/models/retrain	Retrain status / start retraining	Staff / Admin
GET / PUT	/api/settings/risk-policy	Read / update thresholds	Staff / Admin
GET	/api/settings/integrations	PayPal, Groq, database, rate-limiter status	Staff
GET	/api/demo/scenarios	Demo scenario definitions	Staff
POST	/api/demo/scenarios/{A,B,C}/run	Run a scenario through the risk pipeline (no PayPal order)	Staff
GET / POST	/api/demo/simulator	Simulator status / start or stop	Staff

Method	Route	Purpose	Auth
GET	/api/shop/products, /api/shop/products/{id}, /api/shop/demo-presets	Storefront data	Public
GET	/api/health	Liveness and integration flags	Public
WS	/ws/transactions?token=	Live events	Staff token

11.4 Request and response examples

Create order (`POST /api/paypal/create-order`, header `Idempotency-Key: 3f6c...`):

```
{"items": [{"product_id": "PRD_005", "quantity": 2}],
 "device_fingerprint": "fp-ava-macbook-air",
 "geo": null}
```

Response (shape from `order_view`; values from the test suite, which uses the mock PayPal transport):

```
{"paypal_order_id": "5O190100001TEST", "transaction_id": "TXN_...",
 "amount": 96.0, "currency": "USD", "paypal_status": "CREATED",
 "sentinel_status": "AWAITING_BUYER_APPROVAL", "capture_state": "NONE",
 "capture_id": null, "capture_status": null, "risk_level": null, ...}
```

Structured risk result (`POST /api/risk/assess`, abridged):

```
{"transaction_id": "TXN_563TGV4IBG", "fraud_probability": 0.0661,
 "anomaly_score": 0.954, "risk_level": "MEDIUM", "model_version": "v1",
 "model_algorithm": "random_forest",
 "top_risk_factors": ["Transaction amount $1,849.00",
   "First time this customer has used this device",
   "Amount is 43.4x the customer's historical average"],
 "recommended_action": "ADDITIONAL_VERIFICATION",
 "policy_notes": ["Model score 0.066 (random_forest v1) -> LOW",
   "Amount is 43.4x customer average (>= 10.0x) -> at least MEDIUM"]}
```

Reviewer decision (`POST /api/reviews/{txn}/decision`):

```
{"decision": "REJECT",
 "reason": "Shared emulator device with 4 ring accounts and flagged mule destination (E5, E6).",
 "password": "********", "confirm_transaction_id": "TXN_56SIN7VDWC"}
```

Common errors: 401 not authenticated, invalid token or re-authentication failed; 403 wrong role or not the order owner; 404 not found; 409 wrong state (for example not approved by buyer, already decided, amount mismatch, policy violation); 422 validation error (structured `errors` list); 429 rate limit; 502 PayPal error with `debug_id`; 503 PayPal not configured.

PART 12

Security and Reliability

Area	Implementation
Environment variables	All secrets come from environment / `.env` (`pydantic-settings`); `.env` is git-ignored; `.env.example` has blank values; blank values fall back to safe defaults.
Credential management	PayPal secret is used only server-side for OAuth; the client ID is the only PayPal value sent to the browser.
Authentication	bcrypt password hashes; HS256 JWT with expiry (default 480 minutes); failed logins audited.
Role-based access	FastAPI dependencies `require_staff`, `require_reviewer`, `require_admin`; shoppers cannot reach staff endpoints; staff cannot create checkouts.

Area	Implementation
Step-up for decisions	Password re-entry, typed transaction-ID confirmation, at least 20-character justification to approve HIGH/CRITICAL, ADMIN required for escalated cases.
Payment idempotency	Idempotency-Key and PayPal-Request-Id; compare-and-set capture lock; reconciliation of already-captured orders.
Payment ownership	Every order endpoint checks that the caller owns the order (or is staff where allowed).
Webhook verification	PayPal verification API with `PAYPAL_WEBHOOK_ID`; duplicate event IDs ignored.
Input validation	Pydantic models with ID patterns (`^TXN_\w+$` etc.), ranges and lengths; server-side pricing; quantity 1-10; single-seller carts; 256 KB webhook limit.
Rate limiting	Redis-backed when configured, otherwise in-memory; stricter limits on login, create-order and capture.
Audit logging	Every row stores SHA-256(previous hash + content); `/api/audit/verify` detects modification; tested by tampering a row.
Database integrity	Foreign keys, check constraints, unique keys, indexes; SQLite runs with `PRAGMA foreign_keys=ON`.
Error handling	Generic 500 responses; structured 422 errors; PayPal debug IDs preserved.
Secure logging	A logging filter redacts bearer tokens, client secrets and access tokens.
HTTP headers	`X-Content-Type-Options: nosniff`, `X-Frame-Options: DENY`, `Referrer-Policy: no-referrer`; explicit CORS allow-list.
Synthetic data privacy	All people, merchants and devices are generated; device fingerprints are random per-browser IDs.

Current security limitations: JWT is stored in `sessionStorage` (vulnerable if the page has an XSS flaw; an httpOnly cookie would be stronger); no multi-factor authentication; no formal security review or penetration test; the demo password is shared and published for judging; there are no rate-limit or WebSocket protections for distributed deployments beyond Redis counters.

PART 13

User Interface and Product Workflow

All screenshots in this part were captured from the running application on the development machine (1440 x 900 viewport). PayPal and Groq were not configured at the time, which is visible on some screens.

13.1 Sign-in

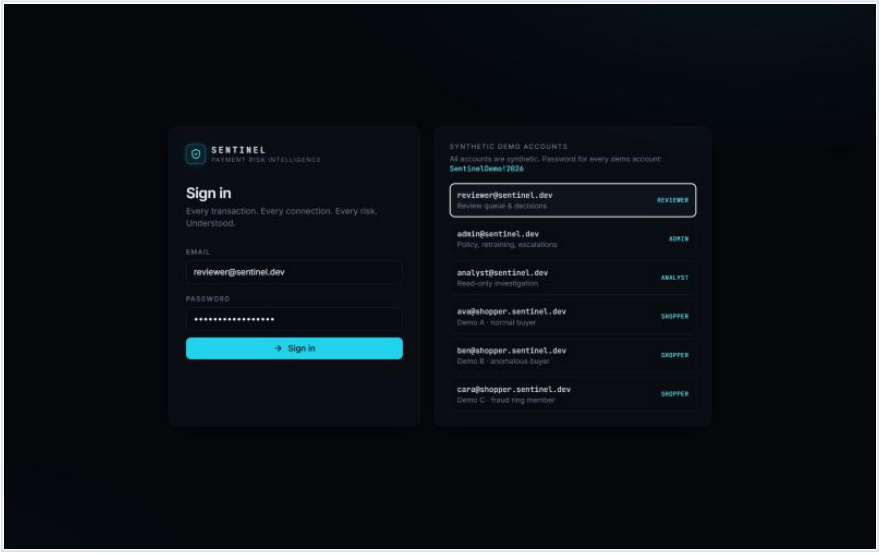


Figure 9: Sign-in page with the list of synthetic demo accounts.

A single sign-in page serves staff and shoppers. Clicking a demo account fills the form. After sign-in, shoppers go to the storefront and staff to the console.

13.2 Customer marketplace (SENTINEL MERCHANT)

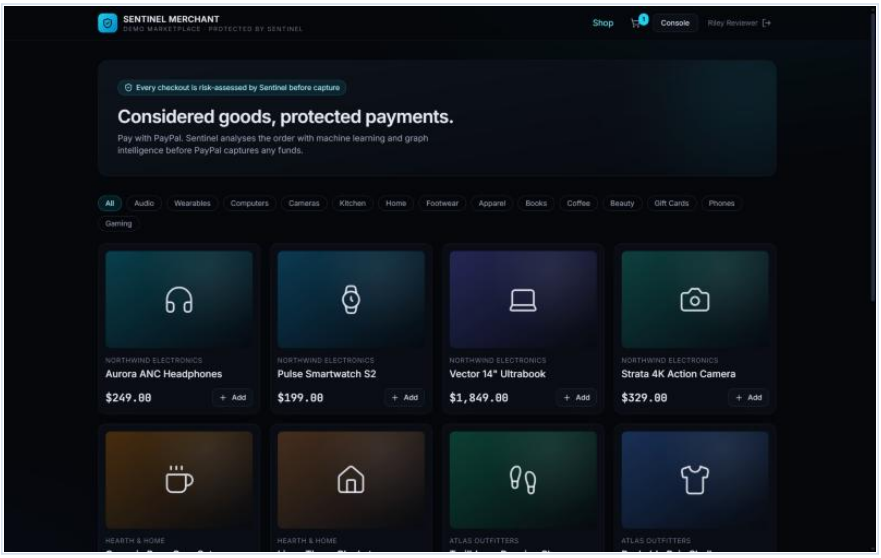


Figure 10: Storefront catalogue with category filters.

Shoppers browse 16 synthetic products from synthetic sellers and filter by category. The banner states that every checkout is risk-assessed before capture.

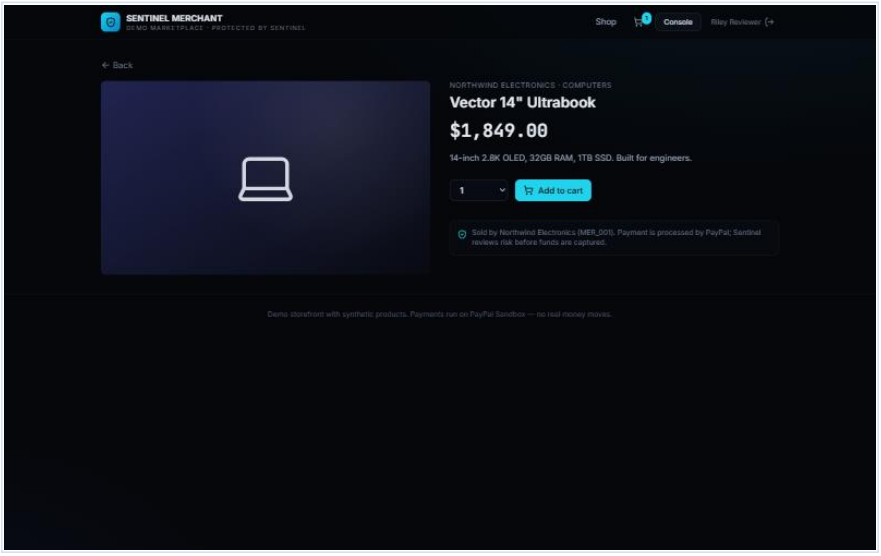


Figure 11: Product detail page.

The product page shows the seller, price and description, and explains that PayPal processes the payment while Sentinel reviews risk before funds are captured. If a shopper adds an item from a different seller, the cart refuses and explains that checkout is per seller.

13.3 Shopping cart and PayPal checkout

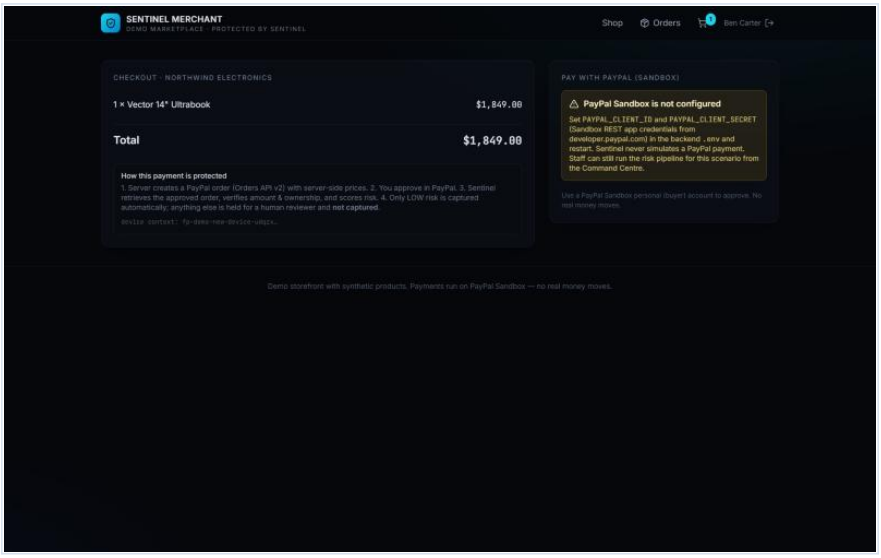


Figure 12: Checkout for Demo B with PayPal not configured: the page explains the setup instead of faking a payment.

The checkout summarises the order, explains the four protection steps and shows the device context. When PayPal is configured, PayPal Buttons appear here. When it is not, as in this screenshot, the page says so explicitly. After approval, a result screen shows **PayPal status** (order, capture, capture ID) and **Sentinel status** (decision, risk level, investigation) side by side.

13.4 Command Centre



Figure 13: Command Centre with KPIs, 30-day volume by risk level, and the demonstration launcher.

The landing page for staff. KPI tiles: transactions analysed, suspicious, critical, awaiting review, PayPal captured, rejected and average processing time. Charts: 30-day volume stacked by risk level, risk distribution, fraud-probability histogram, fraud trend, PayPal and Sentinel status distributions, network summary, model performance and a live event stream. Every number comes from `GET /api/dashboard/metrics`. The demonstration panel runs scenarios A, B and C and starts or stops the synthetic traffic simulator.

13.5 Live Transactions

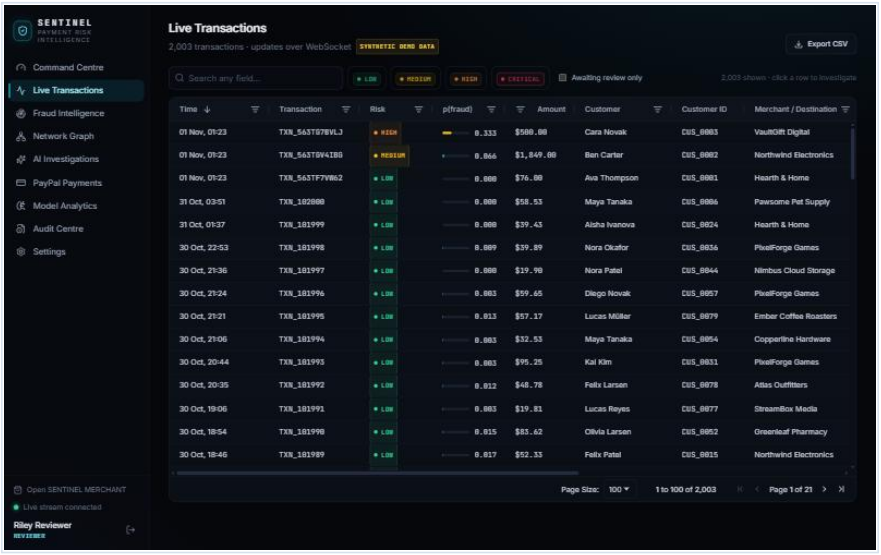


Figure 14: Live Transactions grid (AG Grid) with risk chips, quick search and pagination.

An AG Grid table of all transactions with column filters, sorting, quick search, risk-level chips, an "awaiting review only" toggle and CSV export. Clicking a row opens the investigation workspace.

13.6 Fraud Intelligence

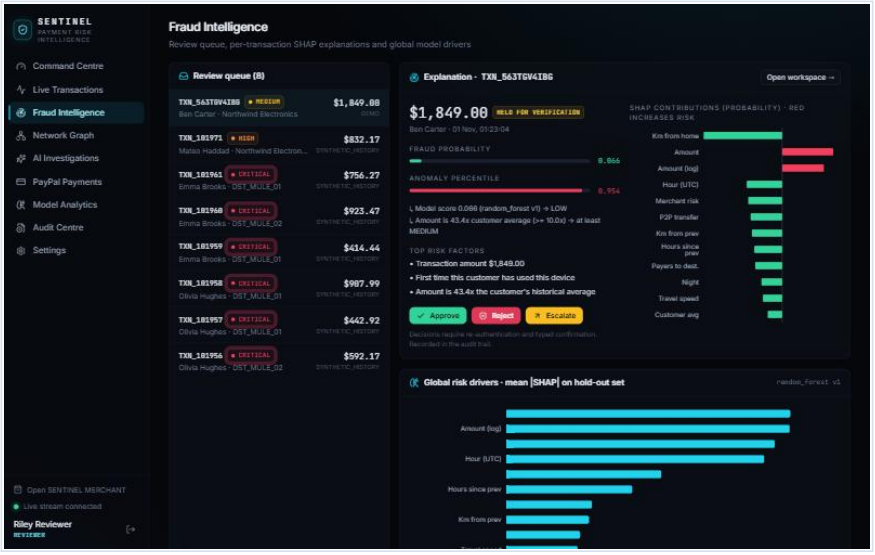


Figure 15: Fraud Intelligence: review queue, per-transaction explanation and decision buttons.

The review queue lists every payment awaiting a decision. Selecting one shows its fraud probability, anomaly percentile, policy notes, top risk factors and a SHAP chart, plus the Approve / Reject / Escalate buttons. Below, a global chart ranks the model's features by mean absolute SHAP value.

13.7 Network Graph

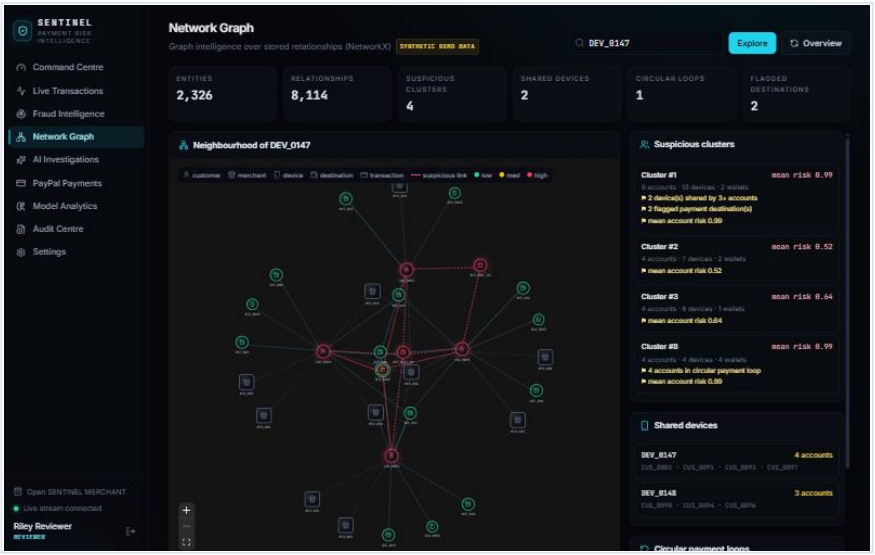


Figure 16: Neighbourhood of device DEV_0147: red dashed edges are suspicious links.

The Network Graph page starts with the overview of suspicious clusters (shown in Part 8) and lets a user search any customer, device or destination ID. Node colour shows risk; icons show type; red animated edges carry evidence. Hovering a node shows its details; clicking a node or edge opens an inspector with evidence and links to the underlying transactions. Side panels list suspicious clusters, shared devices, circular loops and flagged destinations.

13.8 AI Investigation workspace

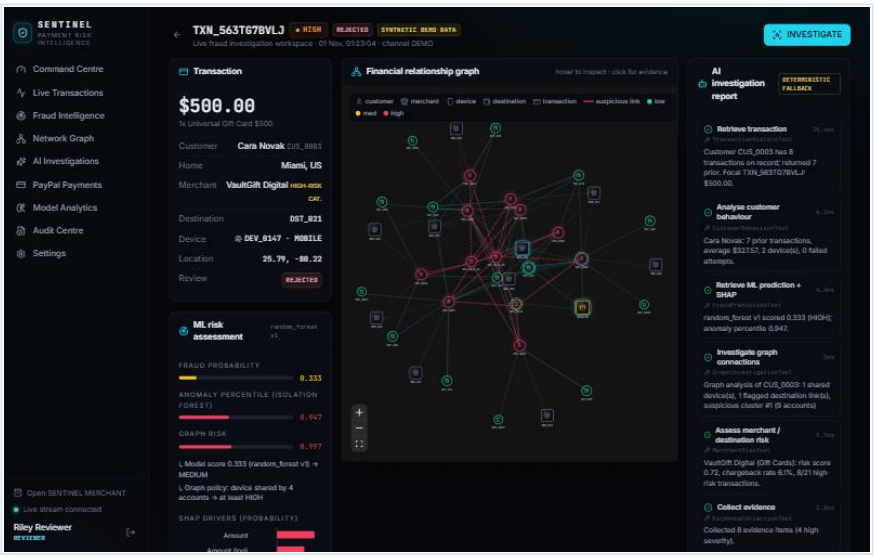


Figure 17: Investigation workspace: transaction (left), relationship graph (centre), AI report steps (right).

The signature screen. Left: transaction details and ML assessment. Centre: the relationship graph around the transaction. Right: the AI investigation with every step, its tool, duration and summary. The INVESTIGATE button starts a new run.

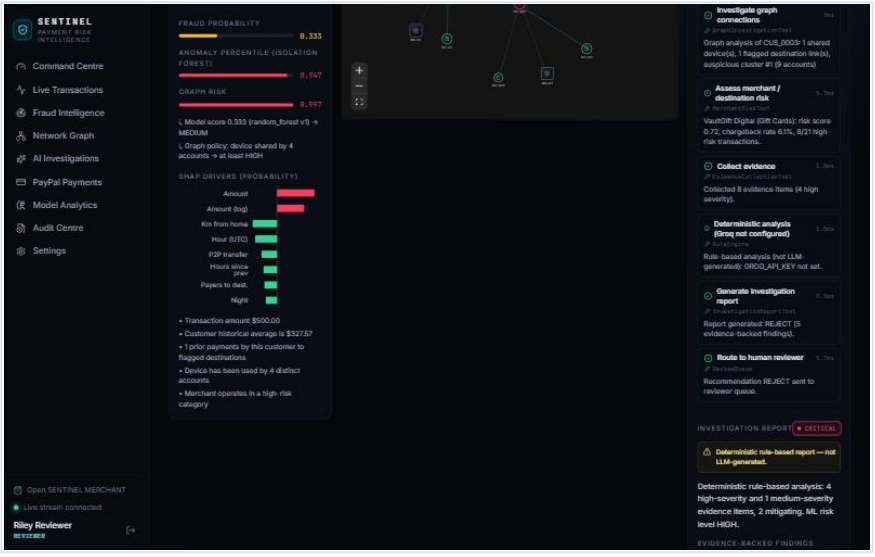


Figure 18: Workspace (scrolled): SHAP drivers on the left; agent steps and the start of the report on the right.

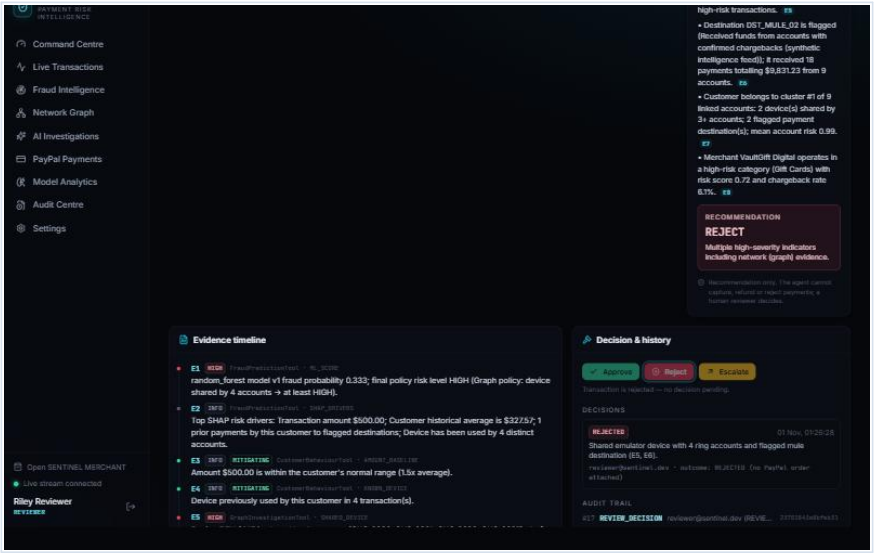


Figure 19: Workspace (bottom): evidence-cited findings, the REJECT recommendation, the evidence timeline and the decision history.

The bottom panels show the evidence timeline (E1…En with severity and mitigating flags), the decision buttons, the PayPal order block when one exists, recorded decisions and the audit trail with hash prefixes.

13.9 AI Investigations list

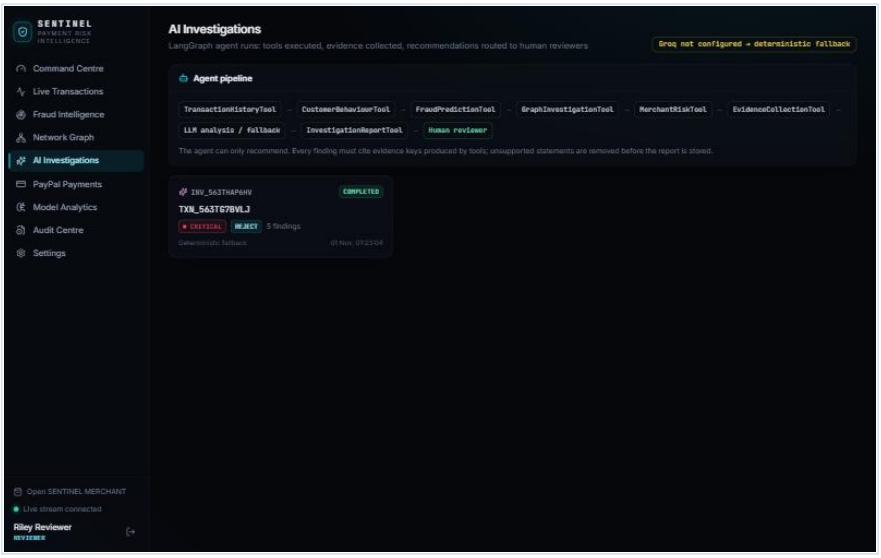


Figure 20: AI Investigations: agent pipeline and completed runs.

Shows the agent pipeline, the current mode (LLM or deterministic fallback) and every investigation with its recommendation and number of findings.

13.10 PayPal Payments

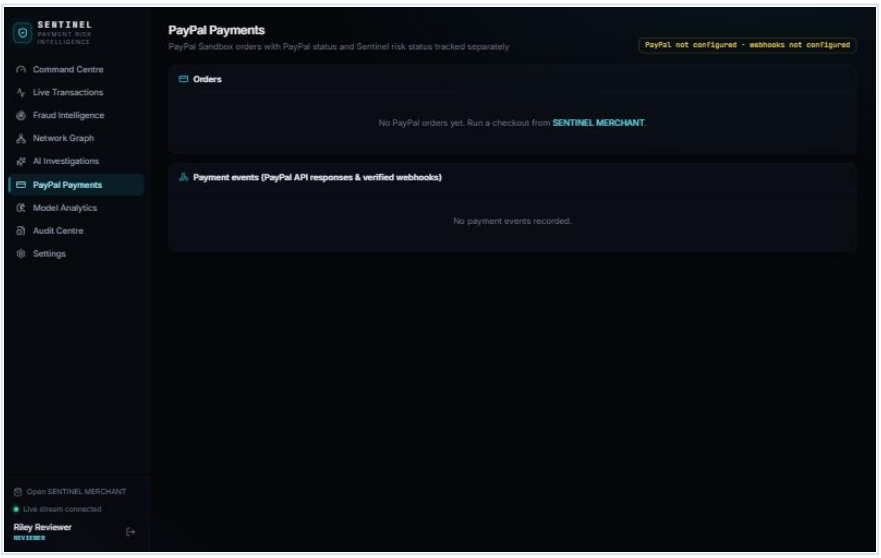


Figure 21: PayPal Payments page: empty because PayPal was not configured during development.

When PayPal is configured, this page lists every order with PayPal status, capture status and capture ID, Sentinel status and risk, plus a "Sync" button that refreshes from PayPal. Below, every PayPal API response and verified webhook is listed with HTTP status and debug ID. It is empty in this screenshot because no live Sandbox orders were created.

13.11 Model Analytics

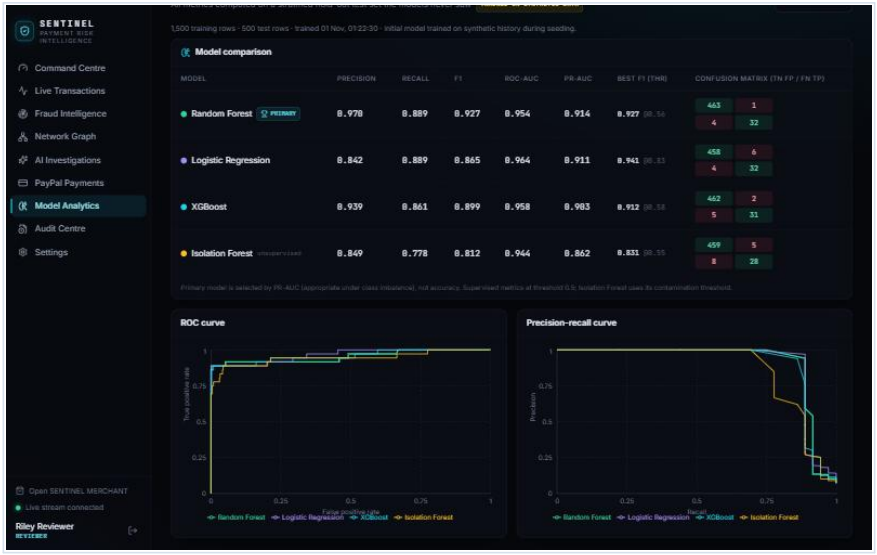


Figure 22: Model Analytics: hold-out metrics, confusion matrices, ROC and precision-recall curves.

Compares the four models, marks the primary, shows confusion matrices and ROC / PR curves, and lets an ADMIN retrain.

13.12 Audit Centre and Settings

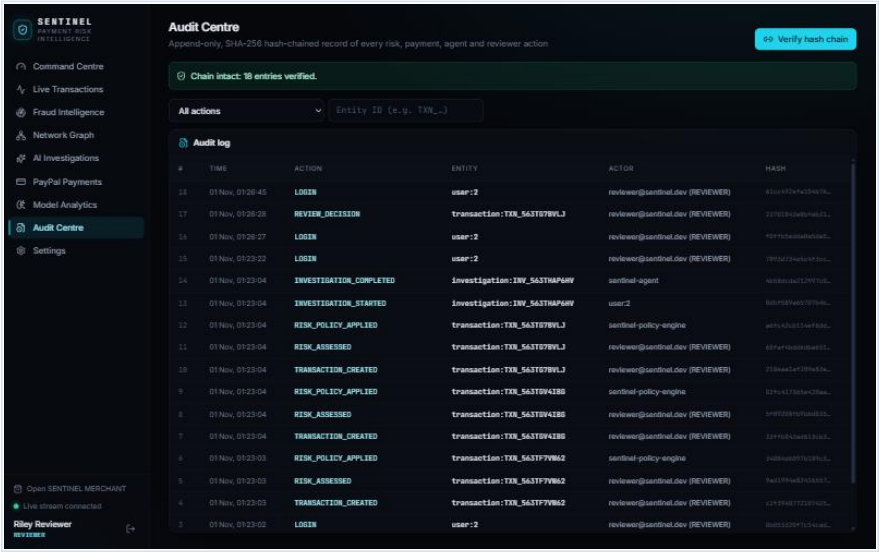


Figure 23: Audit Centre: hash-chained log of every action.

The audit log can be filtered by action or entity ID. "Verify hash chain" recomputes every hash; at capture time it reported "Chain intact: 18 entries verified".

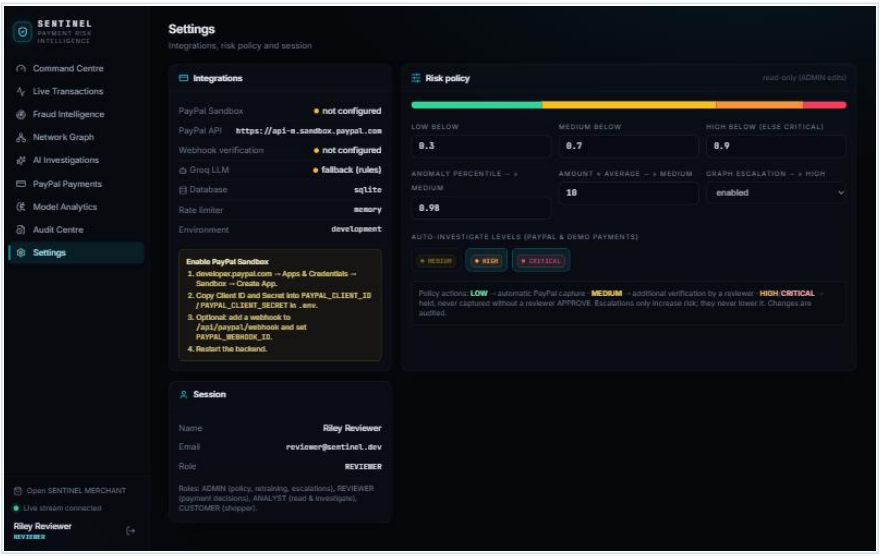


Figure 24: Settings: integration status and the risk policy editor.

Settings shows the PayPal, webhook, Groq, database and rate-limiter status, setup steps for PayPal, and the risk policy (editable by ADMIN; all changes are audited).

PART 14

Demonstration Scenarios

Scenarios are launched from the Command Centre. **"Run risk pipeline"** sends a DEMO transaction through the real risk engine without creating a PayPal order. **"PayPal checkout"** sets the shopper persona and device and opens the real checkout (requires Sandbox credentials). The outputs below are from actual runs of "Run risk pipeline" on the development machine.

Scenario A: Normal payment

Aspect	Detail
Initial conditions	Ava Thompson (CUS_0001), San Francisco, 22 prior purchases averaging about $55, always on her laptop.
User action	Buys a Ceramic Pour-Over Set ($48) and Single-Origin Sampler ($28): $76.00, same laptop, at home.
Backend processing	Transaction created, 26 features computed, prediction stored, policy applied.
ML analysis	Fraud probability 0.000, LOW. No escalation rules triggered.
Graph analysis	No shared devices, cycles or flagged destinations.
AI investigation	Not triggered (LOW).
Payment workflow	Status CLEARED. Through PayPal checkout, Sentinel would call capture automatically.
Final result	Cleared for capture. Audit entries: TRANSACTION_CREATED, RISK_ASSESSED, RISK_POLICY_APPLIED.

Scenario B: Unusual transaction

Aspect	Detail
Initial conditions	Ben Carter (CUS_0002), Chicago, 20 prior purchases averaging about $38.
User action	Buys a Vector 14" Ultrabook, $1,849.00, from a device he has never used.
ML analysis	Fraud probability 0.066 (would be LOW on its own). Anomaly percentile 0.954. Top SHAP factors: transaction amount $1,849.00; amount is 43.4x the customer's historical average; first time this customer has used this device.
Policy	"Amount is 43.4x customer average (>= 10.0x) -> at least MEDIUM".
Graph analysis	Nothing suspicious.
AI investigation	Not auto-triggered at MEDIUM (configurable); can be started with INVESTIGATE.
Payment workflow	HELD_FOR_VERIFICATION, review status PENDING_REVIEW. Through checkout, PayPal would show APPROVED and not captured.
Final result	Waits in the review queue for a human.

This scenario shows why policy rules complement the model: the model, trained on synthetic data that includes legitimate large purchases, gave a low score, and the explicit amount rule plus the explanation put it in front of a reviewer.

Scenario C: Coordinated synthetic fraud network

Aspect	Detail
Initial conditions	Cara Novak (CUS_0003) is part of a 9-account cluster; the emulator device DEV_0147 is shared by 4 accounts; the cluster sent money to flagged wallets DST_MULE_01 and DST_MULE_02.
User action	Buys a $500 Universal Gift Card (VaultGift Digital, high-risk category) on the emulator device.

Aspect	Detail
ML analysis	Fraud probability 0.333, MEDIUM by threshold. Anomaly percentile 0.947.
Graph analysis	"Graph policy: device shared by 4 accounts -> at least HIGH". Graph risk score 0.997.
AI investigation	Started automatically. 8 evidence items, 4 high severity; deterministic fallback report (Groq not configured); recommendation REJECT; 5 evidence-backed findings (Part 9.7).
Human decision	Reviewer re-authenticated, typed the transaction ID and rejected: "Shared emulator device with 4 ring accounts and flagged mule destination (E5, E6)."
Payment workflow	Transaction REJECTED. Through PayPal checkout the order would remain APPROVED and never be captured.
Audit trail	TRANSACTION_CREATED, RISK_ASSESSED, RISK_POLICY_APPLIED, INVESTIGATION_STARTED, INVESTIGATION_COMPLETED, REVIEW_DECISION.

This is the clearest demonstration of the project's thesis: the transaction looked only moderately risky on its own, and the relationships revealed the fraud.

PART 15

Testing and Results

15.1 Automated test suite

The backend has 25 automated tests (`pytest`) in `backend/tests`. They run against an isolated SQLite database that is seeded, trained and scored from scratch, and a **mock PayPal transport** that emulates Sandbox behaviour (OAuth, Orders v2, capture with `PayPal-Request-Id`, `ORDER_ALREADY_CAPTURED`, webhook verification). The mock is only used by tests.

Latest run: **25 passed, 0 failed (32.9 s)**.

Test	Area	Checks
test_models_trained_with_honest_metrics	ML evaluation	All 4 models trained; metrics in [0,1]; confusion matrix sums to test rows; primary has best PR-AUC
test_ml_prediction_structure	ML prediction	Probability and anomaly in [0,1]; SHAP for every feature; positive top factors; missing values kept as null
test_synthetic_world_is_independent_of_seed_time	Reproducibility	Same data whatever day or time the database is seeded
test_risk_classification_policy	Risk classification	Threshold mapping; escalation never lowers risk
test_transaction_creation_and_assessment	API	401 unauthenticated, 403 shopper; risk payload; CLEARED status; audit entry; 422 on negative amount
test_anomalous_amount_escalates	Risk policy	Demo B is MEDIUM or higher and held; amount factor present
test_graph_detects_fraud_structures	Graph	DEV_0147 shared; 4-account cycle; both mule wallets; clean customer has no signals; evidence on edges; 404 unknown entity
test_incremental_graph_matches_full_rebuild	Graph	In-place graph updates match a full rebuild from the database
test_agent_tools_return_real_records	AI tools	Tools return real records; ToolError on unknown IDs; report tool drops invented and uncited findings; invalid recommendation normalised
test_investigation_workflow_deterministic_fallback	Agent workflow	All 7 tools run; labelled fallback; findings cite real keys; agent does not move money

Test	Area	Checks
test_rbac_and_auth	Security	401 / 403 rules; forged token rejected; admin-only policy update
test_audit_chain_detects_tampering	Audit	Chain valid; modified row detected at the right ID; valid again after restore
test_investigations_persisted	Persistence	Completed investigation has evidence and the "recommendation only" notice
test_create_order_uses_server_side_prices	Payment	PayPal status CREATED; server price used
test_create_order_idempotency_key_replays	Idempotency	Same key returns same order; no second PayPal order
test_cart_validation	Input validation	Unknown product, mixed sellers, quantity 99 rejected
test_staff_cannot_create_checkout	RBAC	403 for staff
test_capture_requires_buyer_approval	Payment state	409 when order not approved
test_low_risk_payment_is_captured_once	Capture, duplicates	LOW captured; replays never call PayPal capture again (exactly 1 call)
test_payment_ownership_enforced	Ownership	Another shopper gets 403
test_amount_mismatch_blocks_capture	Integrity	Tampered amount gives 409; no capture
test_high_risk_payment_cannot_bypass_review	Policy	HIGH held, APPROVED not captured, investigation started; direct capture refused (409)
test_reviewer_decision_controls	Review	Shopper 403, wrong password 401, wrong confirmation 422, short reason 422, escalation requires ADMIN, approve captures once, repeat 409
test_reject_decision_never_captures	Review	REJECT leaves PayPal order APPROVED, uncaptured
test_webhook_signature_verification_and_dedupe	Webhooks	Forged or missing signature 400; valid processed; duplicate ignored

15.2 Other verification performed

Check	Result
Frontend type check (`tsc -b`)	Passed
Frontend production build (`vite build`)	Passed
Manual UI walkthrough in a browser	All 9 console sections, the investigation workspace, storefront, checkout (unconfigured) and a reviewer decision exercised
PostgreSQL DDL generation (`alembic upgrade head --sql`)	18 tables generated for the PostgreSQL dialect
ML evaluation	Hold-out metrics in Part 7.6

15.3 Not yet verified

Item	Reason
Live PayPal Sandbox order, approval and capture	Sandbox credentials were not available on the build machine
Live webhooks	Requires a public URL and webhook ID
Groq LLM investigation	No API key; fallback path tested instead
Docker Compose and PostgreSQL at runtime	Docker daemon was not running on the build machine
Render deployment	Not deployed

Item	Reason
Load and performance testing	Not performed
Frontend automated tests	None written

PART 16

Innovation and Differentiation

Compared with	What Sentinel adds
Basic fraud detection (a single score)	Combines a supervised score, an anomaly percentile and graph signals; explains each payment with SHAP and policy notes; holds payments instead of auto-declining.
Traditional rule-based systems	Rules are kept, but as transparent escalations on top of learned models; thresholds are configurable and audited.
Basic AI chatbots	The agent works on real records through tools, must cite evidence, has a validator that removes unsupported claims, and cannot take irreversible actions.
Standalone payment apps	Risk decisions are enforced at PayPal's capture step with idempotency, ownership checks and bypass protection.

The combination is the contribution: PayPal payment infrastructure, ML risk prediction, graph intelligence, an autonomous but bounded investigator, explainable recommendations and human-supervised decisions in one workflow.

Limits of these claims: Sentinel is a hackathon prototype. Commercial fraud platforms already combine many of these techniques at far larger scale with real data. We make no claim that Sentinel is the first or the best system of its kind, or that its accuracy would hold on real payments.

PART 17

Limitations and Future Scope

17.1 Current limitations (implemented system)

- **Synthetic training data.** Patterns were designed by the author; some features (for example hour of day) are strong because of how the generator works.
- **No production validation.** Live PayPal Sandbox and Groq paths are implemented but were not executed during development.
- **Uncalibrated probabilities.** Class weighting shifts scores; they are risk scores rather than calibrated probabilities.
- **Single-process design.** WebSocket fan-out, feature state and graph cache live in one process.
- **CAPTURE intent.** Held orders must be decided within PayPal's approval window.
- **Simplified context.** Device identity is a random browser ID; checkout location defaults to the customer's home city unless a demo provides one.
- **In-sample history scores.** Historical transactions are scored by a model that was trained on most of them; only hold-out metrics measure quality.
- **No frontend tests**, no load testing and no formal security review.

17.2 Future scope (not implemented)

Area	Proposed improvement
Data	Train and validate on real or public fraud datasets; time-based (out-of-time) validation.
Calibration	Isotonic or Platt calibration; cost-based threshold selection.
Monitoring	Feature drift, score drift and alerting; champion/challenger models.
Graph	Community detection (Louvain / Leiden), graph embeddings, graph neural networks, a graph database for scale.
Scale	Redis pub/sub or a message queue for events, a shared feature store, horizontal API scaling, streaming ingestion.
PayPal	AUTHORIZE-intent flow (authorise, then capture or void), refunds, disputes API, live webhook subscriptions, Vault.
Identity	Step-up verification for MEDIUM payments (one-time codes), stronger device intelligence.
Security	httpOnly cookie sessions, MFA for reviewers, penetration testing, secrets manager.
AI agent	Live Groq evaluation, prompt and report quality evaluation, multiple-model comparison.
Product	Case management, reviewer workload metrics, customer communication templates.

PART 18

Conclusion

Sentinel demonstrates that payment infrastructure and AI work best together when each does what it is good at. PayPal provides a real, idempotent payment lifecycle with a clear separation between a buyer's approval and the capture of funds. Machine learning estimates risk and, with SHAP, explains it. Graph analysis reveals coordinated behaviour that single-payment models miss, as Scenario C showed. An agentic investigator assembles the case file, but only from evidence that tools produced, and it can only recommend. Humans make the irreversible decisions, and everything is written to a tamper-evident log.

The engineering concepts combined are payment state machines and idempotency, point-in-time feature engineering, imbalanced classification and honest evaluation, explainable AI, graph algorithms, tool-using LLM agents with validation, role-based security and auditability.

The same pattern of "score, connect, investigate, explain, then let a human decide before money moves" could apply to marketplace payouts, peer-to-peer transfers, refund abuse or merchant onboarding. Sentinel is a prototype on synthetic data, but its architecture is a credible starting point for those problems.